nature food

Article

https://doi.org/10.1038/s43016-024-00954-7

Soil Geobacteraceae are the key predictors of neurotoxic methylmercury bioaccumulation in rice

Received: 29 December 2022

Accepted: 5 March 2024

Published online: 11 April 2024

Check for updates

Huan Zhong[1,2,19] ✉, Wenli Tang[1,19], Zizhu Li[1,19], Christian Sonne[3,4] ✉, Su Shiung Lam[5,6], Xiao Zhang[1], Sae Yun Kwon[7], Jörg Rinklebe[8], Luís M. Nunes[9], Ri-Qing Yu[10], Baohua Gu[11], Holger Hintelmann[12], Martin Tsz-Ki Tsui[13,14], Jiating Zhao[15], Xin-Quan Zhou[16], Mengjie Wu[1], Beibei Liu[1], Yunyun Hao[16], Long Chen[17] ✉, Baogang Zhang[18], Wenfeng Tan[16], Xu-Xiang Zhang[1], Hongqiang Ren[1] & Yu-Rong Liu[16] ✉

Contamination of rice by the potent neurotoxin methylmercury (MeHg) originates from microbe-mediated Hg methylation in soils. However, the high diversity of Hg methylating microorganisms in soils hinders the prediction of MeHg formation and challenges the mitigation of MeHg bioaccumulation via regulating soil microbiomes. Here we explored the roles of various cropland microbial communities in MeHg formation in the potentials leading to MeHg accumulation in rice and reveal that Geobacteraceae are the key predictors of MeHg bioaccumulation in paddy soil systems. We characterized Hg methylating microorganisms from 67 cropland ecosystems across 3,600 latitudinal kilometres. The simulations of a rice-paddy biogeochemical model show that MeHg accumulation in rice is 1.3–1.7-fold more sensitive to changes in the relative abundance of Geobacteraceae compared to Hg input, which is recognized as the primary parameter in controlling MeHg exposure. These findings open up a window to predict MeHg formation and accumulation in human food webs, enabling more efficient mitigation of risks to human health through regulations of key soil microbiomes.

Neurotoxic methylmercury (MeHg) poses health risks to living organisms as it accumulates in human food webs by a factor of up to ten million[1]. Long-term dietary MeHg exposure causes detrimental impacts on children's cognition[2] and reduces intelligence quotients (IQ) by 0.6 million points per year in Europe alone thereby leading to an annual economic loss of €8–9 billion (ref. 3). Among crops, rice has orders of magnitude higher MeHg concentrations than others[4] and is a major dietary source of MeHg, especially in Asia, accounting for up to 96% of the total MeHg exposure in inland areas[5]. The health risk of MeHg in rice is notable even in countries where rice is not a staple food. For instance, the intake of baby cereals leads to a daily intake of MeHg up to 1.2-fold of the reference dose set by the US Environmental Protection Agency (EPA; 0.1 μg kg^{-1} d^{-1}) (ref. 6).

MeHg in rice originates from microbe-mediated Hg methylation in soils[7], whereas the high diversity of putative methylating microorganisms, that is, those containing Hg methylation gene pair *hgcAB*[8], hinders the attempts to mitigate MeHg formation via regulating key microbial methylators. Identifying the key and universally prevailing microbial drivers of MeHg formation in croplands is challenging, considering the presence of a wide diversity of microbial methylators with highly

A full list of affiliations appears at the end of the paper. ✉e-mail: zhonghuan@nju.edu.cn; cs@bios.au.dk; chenlong@geo.ecnu.edu.cn; yrliu@mail.hzau.edu.cn

distinct capacities of Hg methylation (0.6 to 75%) (ref. 9). It also remains unknown whether all *hgcA*-containing microbes contribute to MeHg production, as inconsistent correlations were documented between *hgcA* abundance and MeHg concentration[10]. In addition, the prevalence of Hg methylating microorganisms is co-regulated by soil properties such as organic matter, redox potential and soil C/N ratio[10,11], as higher abundances of *hgcA* could be found in soils either with higher organic matter contents or under anaerobic conditions. These complex effects lead to high spatial heterogeneity of methylating microorganisms in cropland soils[12], where the Hg methylating communities are further impacted by farming activities[13], discouraging efforts to identify the universally prevailing microbial drivers in croplands. To tackle the challenge, it is essential to conduct large-scale assessments of the diversity and the roles of Hg methylating microorganisms in different soils with varied properties before credible conclusions on the microbial drivers behind MeHg formation are drawn. However, despite a previous study that has reported Hg methylating communities in some Hg-mining areas[12], such large-scale field evidence is still lacking.

Here we explore the roles of various cropland microbial communities in MeHg formation on a large scale and particularly in the potentials leading to MeHg accumulation in rice, with the attempt to seek the possibility of mitigating microbial MeHg formation and thus dietary MeHg exposure. First, we collect fresh paddy soil samples covering croplands in tropic, subtropic, warm-temperate and mid-temperate zones in the Northern Hemisphere (geographic information is listed in Supplementary Table 1 and depicted in Extended Data Fig. 1) and identify the key Hg methylating microorganisms by characterizing the microbial communities. We then quantitatively compare the relative importance of the identified key methylating microorganisms and other factors (for example, Hg input via atmospheric deposition and irrigation) in controlling MeHg concentrations in rice by developing a refined rice-paddy biogeochemical model (RPBM) based on our large-scale field measurements. Consequently, we explore the possibility of mitigating MeHg exposure via regulating the key methylating microorganisms by conducting model simulation and cost estimation. Results obtained from this study improve the ability to predict MeHg accumulation in rice and allow the development of more efficient strategies for mitigating the health risk of MeHg in crops.

Results

Highest Geobacteraceae proportion in methylating communities

We reveal that the *hgcA*-containing iron-reducing bacteria (FeRB) Geobacteraceae are the most abundant methylating assemblies in paddy soils on a continental scale. Geobacteraceae, an important family in Deltaproteobacteria, is a bacterial taxon that dominates Fe reduction in paddy soils[14] and has been implicated in Hg methylation[12,15]. The relative abundance of Geobacteraceae across China is averagely 50%, showing the highest proportion in eastern and northeastern China (67.5%; Fig. 1a,b) and the lowest in southwestern China (34.7%). The geographic distribution of Geobacteraceae in paddy soils is correlated with environmental factors, including temperature, soil Se concentration, organic matter content and so on (Supplementary Discussion 1 and Supplementary Table 2). Notably, the highest abundance of Geobacteraceae, among all *hgcA*-containing Hg methylating microorganisms, is detected in soils during the whole cycle of rice cultivation and in either non-rhizospheric or rhizospheric soils (Extended Data Fig. 2). Other identified Hg methylating microbes, including methanogens (predominately related to Methanomassiliicoccaceae, Methanosarcinaceae and Methanoregulaceae) and sulfate-reducing bacteria (SRB; dominated by Desulfovibrionaceae, Desulfobulbaceae and Desulfobacteraceae) have lower relative abundances, only accounting for an average of 13.6% and 6.7% of the total detected *hgcA* sequences, respectively (Fig. 1a,c).

Dominant role of Geobacteraceae in MeHg production

The relative abundances of methylating microorganisms do not necessarily reflect the microbial abilities to produce MeHg, considering that no significant relationship between Hg methylation potential (the ratio of soil MeHg to THg, MeHg/THg, %, Fig. 1d; addressed in Supplementary Discussion 2) and the abundance of methylation gene *hgcA* is found in this (Extended Data Fig. 3) and previous studies[16,17]. It is, therefore, essential to compare the relative contributions of Geobacteraceae with other Hg methylating taxa to MeHg formation on a large scale. To achieve this goal, we examine the relative contributions of major methylating microbes by conducting metabolism assays. We find that the inhibition of Fe(III) reduction in soils (by selenium/Se; Methods and Supplementary Discussion 3) leads to significant reductions in MeHg formation and the decreases in FeRB activities (indicated by HCl-extractable Fe(II)[18]) are significantly correlated with the decreases in MeHg formation (Fig. 2a). Additionally, HCl-extractable Fe(II) levels are significantly correlated with MeHg concentrations in soils irrespective of the inhibitions of SRB and/or methanogens (Fig. 2b). These provide hints for the major contribution of FeRB to MeHg formation (Supplementary Discussion 3). Although it remains unknown how Fe reduction is involved in MeHg production, we propose two possible pathways: (1) the *hgcA*-containing FeRB directly involve in Hg methylation as Hg microbial methylators and (2) FeRB indirectly impact MeHg production via regulating Fe reduction, inorganic Hg (IHg) binding and thus IHg mobility[19]. It is important to note that Se amendment at a dose (0.4 mg kg^{-1}) comparable to background Se concentration in soils (0.39 mg kg^{-1} for 67 studied soils) does not reduce but even increases IHg mobility (Extended Data Fig. 4), although IHg immobilization is considered the mechanism of Hg–Se antagonism in soil (Supplementary Discussion 3). In contrast, the inhibition of sulfate reduction (Extended Data Fig. 5) does not reduce the net MeHg production even in soils with relative high SRB abundances (Fig. 2c), implying that SRB have a lower contribution than FeRB to MeHg formation in soils. This is also supported by the lack of a significant relationship between MeHg concentrations and sulfate levels, reflecting SRB activities (Fig. 2d). Similarly, methanogenesis inhibition does not lead to significant changes in MeHg formation, except in soils with high methanogen abundances (Fig. 2c). These results reflect a relatively low contribution of methanogens to net MeHg production in paddy soils, also considering the generally low relative abundance of methanogens (that is, 13.6%; Fig. 1a). The co-inhibition of methanogens and SRB activities does not reduce MeHg concentrations in soils either (Fig. 2c), further suggesting that SRB and methanogens play a relatively minor role in MeHg formation in paddy soils on a large scale.

The dominant role of Geobacteraceae in MeHg formation is further supported by our large-scale field survey, showing a strong connection between Hg methylation and the abundance of Geobacteraceae. There is a robust spatial gradient in MeHg/THg ratio showing the highest methylation in northeast and the lowest in southwest China, which is in line with the spatial distribution of Geobacteraceae (Fig. 1a,b,d). The significantly positive relationship between Hg methylation potential and the relative abundance of Geobacteraceae (Fig. 3a) further supports the crucial role of Geobacteraceae in MeHg formation in cropland soils at a large spatial scale. Compared with Geobacteraceae, the relative abundance of SRB is less strongly correlated with Hg methylation potential (Fig. 3b), though this microbial group is widely recognized as an important group responsible for MeHg formation in other environments[16,20]. Unexpectedly, methanogens and other unclassified *hgcA*-containing microorganisms are negatively related to Hg methylation potential (Fig. 3c,d), and their methylation role could be obscured by their possible involvement in MeHg demethylation in soils[21]. These results from our large-scale assessment further highlight the greater importance of Geobacteraceae than other major methylating microbes in driving MeHg formation in general paddy soil ecosystems.

https://doi.org/10.1038/s43016-024-00954-7

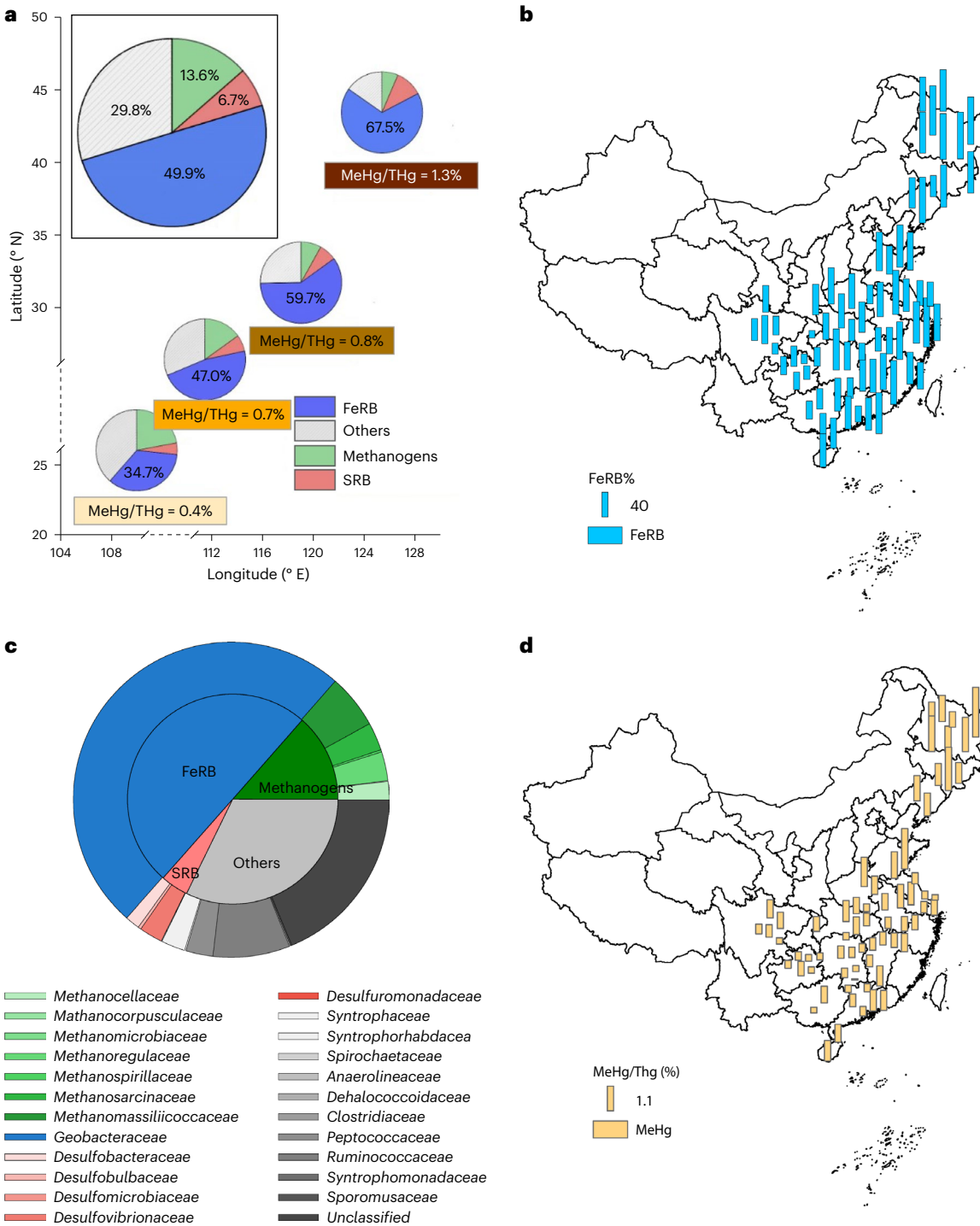


Fig. 1 | The distribution patterns of Hg methylating microbial communities and Hg methylation potentials in rice-paddy soils across China. a–d, This comprises the overall geographic distributions of the relative abundance of Hg methylating microbes (containing *hgcA* gene) and the Hg methylation potential (indicated as the ratio of soil MeHg to soil total Hg, %, reasons given in Supplementary Discussion 2) in different regions of China (**a**), presented as the regional average of latitude and longitude, and each site (**d**), the relative abundance of FeRB in the detected *hgcA* sequences on the provincial scale (**b**) and the national average composition of microbial methylators (**c**). Breaks were set in the *x*- and *y*-axis of **a** to avoid pie overlap due to a higher sample density in this area. Basemaps in **b** and **d** from the Resource and Environment Science and Data Center (https://www.resdc.cn/Datalist1.aspx?FieldTyepID=20,0).

Controlling role of Geobacteraceae in rice MeHg accumulation

The accumulation of MeHg in rice depends not only on microbe-mediated Hg methylation in soil, but also on processes such as Hg input through atmospheric deposition and irrigation, Hg output through evasion and run-off and MeHg absorption by plants (Fig. 4a). Nevertheless, the process dominating MeHg accumulation in rice remains unclear, which hinders the prediction of MeHg risk in rice and the development of strategies to effectively mitigate the risk. Kwon et al.[22] developed a RPBM, a powerful tool for quantifying various biogeochemical processes affecting MeHg accumulation in rice. We refine the RPBM by incorporating the microbial methylation process (Supplementary Methods), which is the impetus for the bioaccumulation and risk of

https://doi.org/10.1038/s43016-024-00954-7

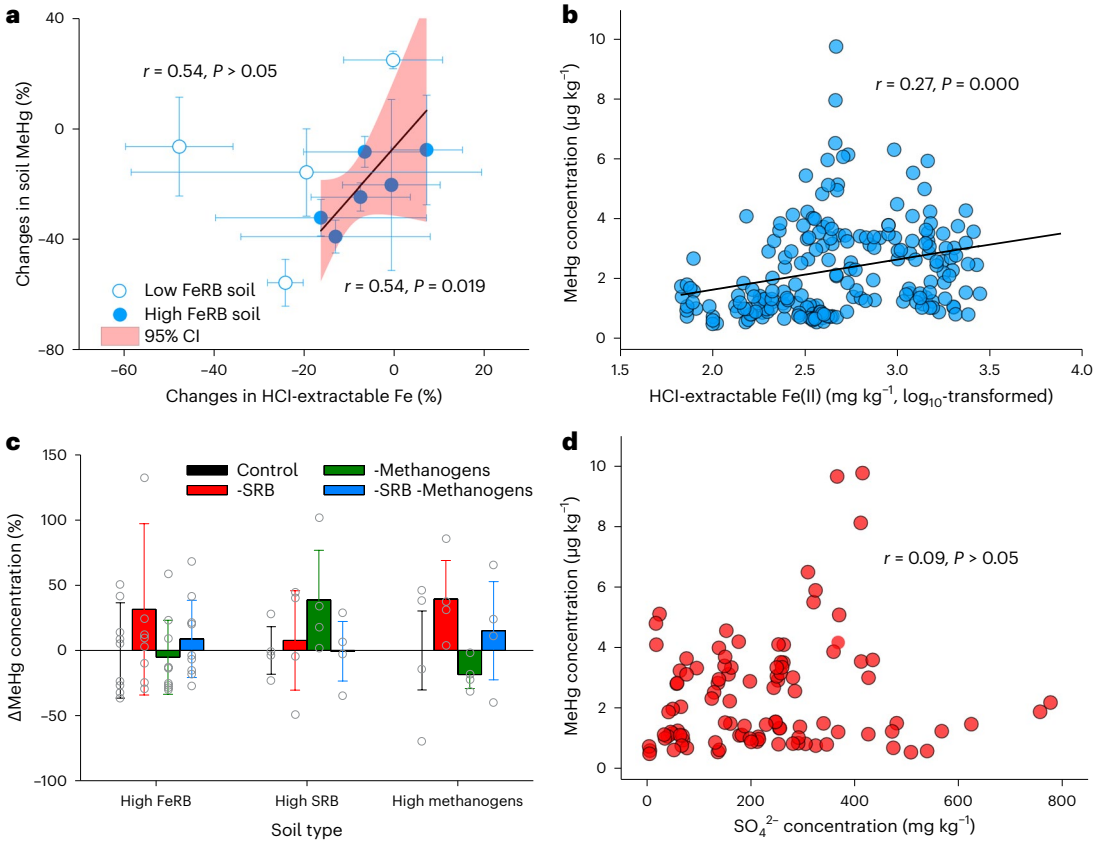


Fig. 2 | Evidence obtained from metabolism assays that supports the dominant role of Geobacteraceae in MeHg formation. a, The relationships between changes in MeHg levels and changes in HCl-extractable Fe(II) in ten soils added with selenium (Se, 0.4 mg kg^{-1}; Assay II). **b**, The relationship between MeHg concentrations and HCl-extractable Fe(II) contents. **c**, Changes in MeHg concentrations. **d**, The relationship between MeHg concentrations and sulfate levels. For panel **a**, data are presented as mean ± standard deviation, $n = 3$ replicates. Data depicted in panels **b**–**d** are obtained from 17 soils with/without SRB and/or methanogens inhibited (Assay I). In panels **b** and **d**, each dot represents a replicate. In panel **c**, ΔMeHg concentration (%) was calculated as the difference in MeHg concentration between the treatment and the control group. Data are presented as mean ± standard deviation, $n = 9$, 4, 4 soils for high FeRB, high SRB and high methanogens soils, respectively. Each dot on the bar represents the average of 3 replicates for a soil. *P* values are obtained based on a two-sided test. Error bands are 95% confidence intervals (CI) of the regression lines.

MeHg. Our improved model shows a good performance in predicting aquatic THg and IHg concentrations (Extended Data Fig. 6), soil THg and MeHg concentrations (Extended Data Fig. 7 and Supplementary Discussion 4) and rice MeHg concentrations across China (Fig. 4b,c). By applying Monte Carlo simulations to the improved RPBM, we find that MeHg concentration in rice is the most sensitive to the MeHg bioaccumulation factors (BAFs) (Fig. 4d). This is attributed to the fact that BAF is a numerical coefficient compiling multiple processes affecting MeHg bioaccumulation, including MeHg adsorption, translocation and potential transformation within plants (Supplementary Discussion 5). Besides BAF, the abundance of *hgcA*-containing Geobacteraceae is the most sensitive parameter in determining MeHg accumulation in rice grains compared with other microbial methylators and Hg input via irrigation and atmospheric deposition. The changes of rice MeHg concentration in response to 10% changes in Geobacteraceae abundance are 3.3–18.2-fold higher than those in response to the same changes in SRB or methanogen abundance (Extended Data Fig. 8), further supporting the stronger control of Geobacteraceae on rice MeHg levels than other Hg methylating microorganisms. The sensitivity of rice MeHg concentration to Geobacteraceae abundance is 1.3- and 1.7-fold higher than the sensitivity to irrigation and atmospheric deposition, respectively (Fig. 4d). These findings are contrary to the common belief that atmospheric Hg deposition controls MeHg accumulation in rice[22]. Similarly, the changes in health risks caused by MeHg exposure, indicated by hazard quotient (HQ) and intelligence quotient decrement (ΔIQ), are more sensitive to the changes in the proportion of Geobacteraceae abundance (7.9%) compared with atmospheric Hg deposition (2.4%; Extended Data Fig. 9). Particularly for soils with abundant Geobacteraceae, the health risks of MeHg are more sensitive to the changes in Geobacteraceae proportion than Hg input. For instance, in Suzhou, Jiangsu Province (JSSZ), a region with thousands of years of history in rice cultivation[23] and a relative Geobacteraceae abundance of 80.1%, reducing Geobacteraceae abundance in soil by 10% leads to an 8.9% reduction in HQ. In contrast, HQ decreases only by 3.5% or 6.5% in response to a 10% reduction in atmospheric Hg deposition or Hg input via irrigation, respectively (Supplementary Table 3). This highlights the role of Geobacteraceae as a predictor of MeHg accumulation in rice and the subsequent risks in areas with high Geobacteraceae abundance, such as northeastern China (with an average Geobacteraceae abundance of 68.4%), a major rice production region in Asia[24].

Mitigating Hg risk via regulating Geobacteraceae in soil

By revealing the dominant role of Geobacteraceae in predicting MeHg accumulation in rice grains, we hypothesize that microbial regulation might be an alternative approach to mitigate MeHg exposure risk. Particularly, our large-scale field data reveal that Se level in the soil is a key factor controlling either the relative abundances of Geobacteraceae or MeHg production in paddy soils (Fig. 5a,b and Supplementary Discussions 1 and 3). The inhibition of Se on Geobacteraceae abundance/activity is further supported by the results of metabolism inhibition

https://doi.org/10.1038/s43016-024-00954-7

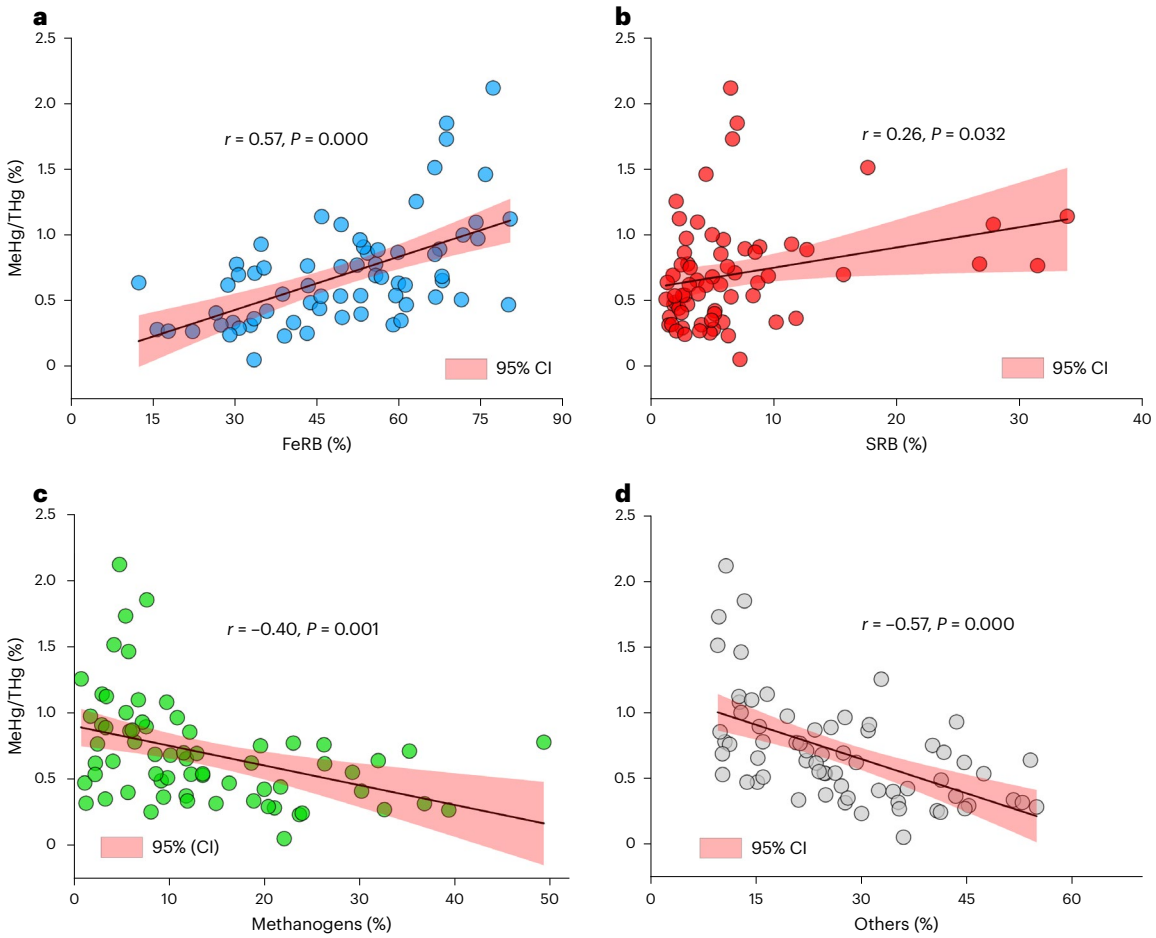


Fig. 3 | Relationships between Hg methylation potential in soil and the relative abundance of different microbial methylators. The ratio of soil MeHg to total Hg (MeHg/THg, %) indicated the potential of Hg methylation. **a**–**d**, The data on the *x* axis indicated the relative abundances of *hgcA*-containing FeRB Geobacteraceae (**a**), *hgcA*-containing SRB (**b**), *hgcA*-containing methanogens (**c**) and other *hgcA*-containing microorganisms (**d**) to the total detected *hgcA* sequences, that is, the proportion of each cluster to the total detected *hgcA* sequences. *P* values are obtained based on a two-sided test. Error bands are 95% confidence intervals of the regression lines.

assays (Fig. 2a) in which the reduction of Fe(III) is inhibited by the amendment of Se at an environmentally relevant dose. By applying the relationship between Se levels and the proportions of Geobacteraceae and the relationship between the proportions of Geobacteraceae and MeHg/THg ratios into RPBM (Supplementary Methods), established via our large-scale field survey, we could quantify the changes in rice MeHg induced by Se application, being commonly and globally used for fertilizing croplands to enhance Se content in foods[25,26]. Results show that there is a significant and positive relationship between the decrease in rice MeHg level and the amount of Se application (Fig. 5c). We further compare the costs between Se fertilization and reducing Hg emissions from coal-fired power plants, a main measure to mitigate MeHg exposure[27]. It is estimated that the total cost is ~US$32.24–40.79 million for the emissions reduction during 2011–2015 in China (Supplementary Methods), whereas the total cost of Se application is US$6.08 million to achieve the same reductions in rice MeHg via Hg emissions control in coal-fired power plants (Supplementary Discussion 6 and Supplementary Table 4). The application of Se to soil results in a minor elevation in soil Se concentration by 0–0.023 mg kg^{-1} and is less likely to cause Se contamination. These results suggest that Se fertilization could be an alternative and effective approach to reduce MeHg accumulation in rice on a large scale, with hidden while evident impacts on mitigating Geobacteraceae-driven MeHg production in soil. However, caution must be exercised when regulating certain microbes in paddy fields, considering that Geobacteraceae is also linked to elemental cycling such as C, N and metals[28]. A comprehensive evaluation of the impacts of microbial regulation on element biogeochemistry and soil fertility, especially the potential ecological impacts, should be conducted before this approach can be promoted to mitigate MeHg risks.

Discussion

We demonstrate that Geobacteraceae, a group of globally ubiquitous bacteria in environments[29], play the dominant role in predicting MeHg formation, accumulation and the associated health risks in paddy soil ecosystems. In contrast to the current knowledge that sulfate-reducing bacteria are the key Hg methylators primarily in mining-contaminated areas[13], our study provides insights and empirical evidence that Geobacteraceae could be the prime cause of MeHg formation and bioaccumulation in the rice-paddy soil system on large spatial scale. Despite a few site-specific reports on the involvement of Geobacteraceae in Hg methylation, for example, in a wastewater treatment plant[15], periphytic biofilms[30] and two Hg-mining sites[12], their universally prevailing role as the key methylating microbes in croplands and critical roles in MeHg bioaccumulation in human food webs have never been established and are largely discounted. In this study, we reveal that MeHg accumulation in rice grains and MeHg exposure risk are more sensitive to Geobacteraceae than to Hg input in areas with high Geobacteraceae abundances (Fig. 4d, Extended Data Fig. 9 and Supplementary Table 3). The identification of the universally prevalent and key microbial drivers of MeHg formation and MeHg accumulation

https://doi.org/10.1038/s43016-024-00954-7

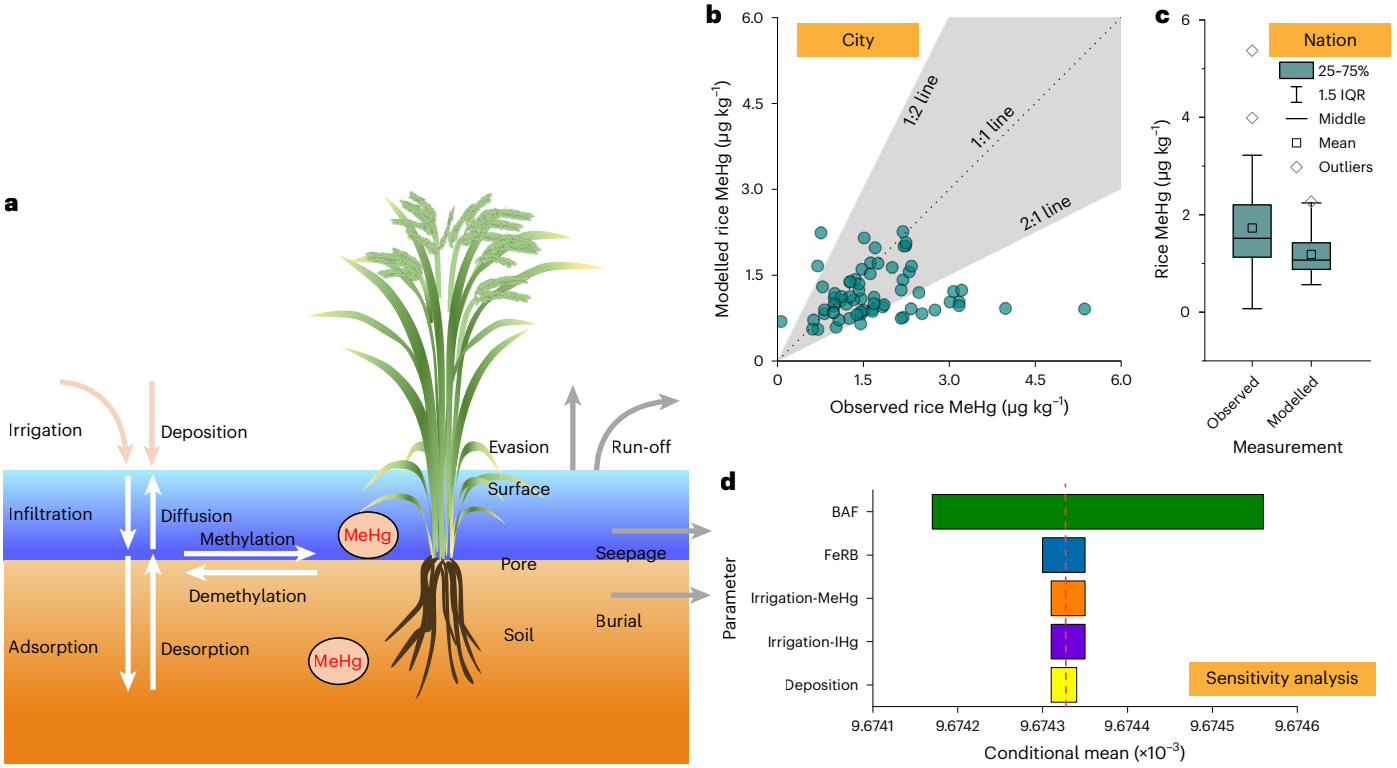


Fig. 4 | The framework of rice-paddy biogeochemical model and its evaluation. a–d, The construction of the modified RPBM (with a depth consistent with soil sampling in the large-scale survey) (**a**), the comparison between modelled rice MeHg concentrations and observed values on the city (**b**) or national scale (**c**) and the results of sensitivity analysis by applying Monte Carlo simulations to RPBM (**d**). For box plot, the box, whisker, line, square and diamond indicate a 25–75% range, 1.5 IQR range, median, mean and outliers, respectively.

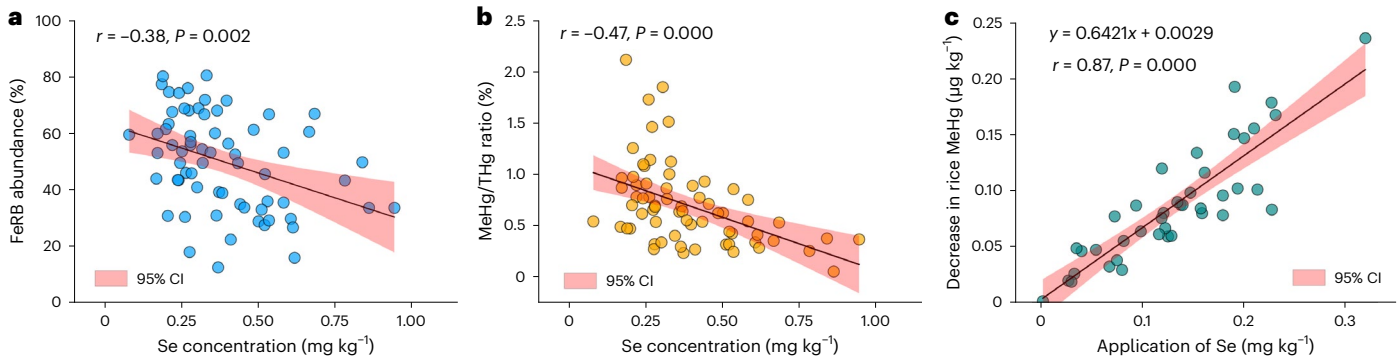


Fig. 5 | Evidence supporting Se fertilization as an alternative approach to effectively mitigate Hg risk. a–c, The correlation between the abundance of *hgcA*-containing FeRB Geobacteraceae and soil Se concentration (**a**) and the correlation between Hg methylation potential (the ratio of soil MeHg to THg, %) and soil Se concentration in the continental survey (**b**); the modelled decreases in rice MeHg induced by Se amendment into soils (**c**). *P* values are obtained based on a two-sided test. Error bands are 95% confidence intervals of the regression lines.

among diverse parameters helps to reduce the uncertainty in predicting MeHg accumulation in crops, which is essential for evaluating the effectiveness of the United Nations Environment Programme (UNEP) *Minamata Convention on Mercury* in mitigating dietary exposure. Geobacteraceae are abundant in Hg methylation hotspots, including wetlands (this study), estuaries[31] and lake sediments[15,32]; therefore, their role in driving MeHg flux into human food webs should also be emphasized, investigated and elucidated.

Microbes have been known for decades as the dominant producers of MeHg[33]. The regulation of methylating microorganisms, however, has rarely been considered as a candidate approach to reduce global MeHg exposure risk, largely because of the highly diverse microbes being involved and their spatial heterogeneity. Our results support that regulating the key methylating microorganisms, that is, Geobacteraceae, is a promising shortcut for mitigating MeHg risk in rice. It is important to note that using Se to regulate Geobacteraceae is still in its infancy. Further studies are warranted to reveal the potential mechanisms, and a comprehensive evaluation is necessary, for example, quantifying the inherent Se concentrations in soils to avoid potential Se contamination and toxicity, monitoring the changes in the ecological functions of Geobacteraceae in paddy soils upon regulation and assessing the long-term effectiveness of microbial regulation in the field. More practical, targeted and economical strategies of microbial regulation are thus possible for reducing the global risk of dietary MeHg exposure.

Furthermore, we show that microbial regulation has advantages in alleviating MeHg risk in risk-sensitive sites, for example, rice-paddy fields. Currently, the global goal to reduce MeHg accumulation in foods is achieved mainly through Hg emissions control, whereas its effectiveness in reducing the exposure risk of MeHg to humans and wildlife is currently difficult to evaluate[34]. This is partly because MeHg accumulation in biota is sometimes inconsistent with the changes in Hg emissions[35]. Our sensitivity analysis confirmed that MeHg accumulation is less sensitive to Hg deposition derived from emissions (Fig. 4d and Extended Data Fig. 9) and a recent trial, in which the changes in Hg health risks and Hg deposition were estimated only ~1/3 and ~1/2 of the variations in Hg emissions, respectively[36]. Meanwhile, the efficacy of Hg emissions control may be undermined by two key factors: the natural Hg emissions accounting for about 50% of the total annual emissions and the mobilization of legacy Hg in response to changing environments. Besides, the protective effects of Hg emissions control on ecosystems are untargeted. It is because the decreases in Hg deposition due to Hg emissions control are found not only in Hg methylation hotspots such as paddy soils, wetlands, lakes and oceans that sustainably supply food for humans but also in areas with low MeHg risks such as deserts. In contrast, for Hg methylation hotspots such as paddy soils, their distributions are highly dispersive, making it difficult to targetedly reduce Hg deposition and the subsequent risks in specific sites via emissions control alone. In addition to effectiveness and pertinence, the cost of Hg emissions reduction keeps increasing since more effective Hg emissions control devices with higher Hg capture efficiencies and costs ranging from 1,841 to 10,356 US$ kg^{-1} Hg (ref. 37) are indispensable due to the obligation of the *Minamata Convention on Mercury*. By contrast, the cost of microbial regulation, for instance, via Se fertilization, could be much lower, as demonstrated above (further discussed in Supplementary Discussion 6). Particularly, microbial regulation in soils with high Geobacteraceae abundances and high methylation potentials, in combination with current measures of Hg immobilization (for example, by using biochar) in areas with low Geobacteraceae abundances, could be effective strategies for point treatment in Hg-contaminated sites. Incorporating these point treatments into the strategic plan (the ongoing Hg emissions reduction) is envisaged to improve the feasibility and cost effectiveness of ensuring food security.

Methods

Field survey and laboratory experiments

Soil sampling. To obtain soils that are representative of soil–rice systems on a national scale, 67 sites of paddy soils (details in Supplementary Table 1 and Extended Data Fig. 1) from 12 provinces in China were collected right after the rice harvest in November 2017. These provinces were selected because they are recognized as primary rice-planting areas, accounting for 82% of the total rice production in China. Meanwhile, these provinces distribute on a large scale, with longitude ranging from 102.74° E to 130.59° E and latitude ranging from 18.26° N to 48.04° N. Such a large-scale sampling area covers tropic, subtropic and temperate zones in the Northern Hemisphere, with various climatic characteristics (for example, annual temperature and precipitation; Supplementary Table 1). This enables the analyses of the impacting factors on both microbial community and Hg methylation potential. Rice paddies are generally drained at the maturation stage in China; therefore, the wet paddy soils sampled right after harvest, with a relatively high mass water content of 40.2%, were similar to those during rice growth. In addition, sampling after harvest is a common approach for analysing functional genes by focusing on genomic DNA residues, considering they are stable in soils for months[38]. Soils were collected at a depth of 1–13 cm because this is a puddling soil layer where rice roots exist and is recognized as a hotspot of Hg methylation[39]. For each sample, five soil columns from different subsites were collected and immediately transported back to the laboratory in ice boxes. These soil samples were mixtures of rhizospheric and non-rhizospheric soils due to the relatively high planting density in China. The soils from the same site were homogenized and divided into subsamples, which were used for the determination of Hg concentrations (total mercury/THg and MeHg, details below; Extended Data Fig. 1b,c), *hgcA* gene abundance (details below[15], *hgcA* community composition, the constants of Hg methylation and MeHg demethylation rates (k_m and k_d, details below[40]) and soil parameters (for example, sulfate/SO_4^{2-}, selenium/Se, total iron, total carbon and total nitrogen and so on). Details of parameter analyses are listed in Supplementary Table 5.

To compare the microbial composition of *hgcA*-containing communities in soils during different stages of rice growth and in rhizospheric and non-rhizospheric soils, additional paddy soil samples were collected from June to October 2020. Our sampling strategies, that is, national scale sampling right after rice harvest and specific site sampling during rice growth, ensure that the samples were representative of various soil types and diverse soil conditions at different rice growth periods. Fewer sampling sites were selected here because we analyse the microbial methylation community based on all existing DNA, which is stable for months in situ[38]. Specifically, six paddy soil samples were collected at the tillering stage from Hubei Province, and two paddy soils were collected at the filling stage in Jiangsu Province, China. Additionally, five groups of paired rhizospheric and non-rhizospheric soils were collected from Hubei Province right after the rice harvest. Rhizospheric soils were collected as the remaining soil particles attached to rice roots after shaking the roots strongly[41]. The non-rhizospheric soils were collected as the shake-off soils.

Metabolism inhibition assays. To provide further evidence of FeRB dominating Hg methylation in paddy soils, we conducted metabolism inhibition assays to assess the respective contributions of major *hgcA*-containing microorganisms to MeHg formation. Three clusters of dominant methylating microbes were studied, that is, SRB, FeRB and methanogens, considering these three clusters have been suggested as important methylating microbes in sediments and soils[9]. Syntrophs are known to be involved in Hg methylation and have been documented to play a dominant role in MeHg production in sulfate- and iron-limited environments[42]. They may have a less notable contribution to MeHg production in paddy soils, which exhibit high sulfate (8–1,073 mg kg^{-1} in the 67 soils, average 152 mg kg^{-1}) and Fe levels (1.1–6.5% in the 67 soils, average 3.6%). No significant correlation was found between Hg methylation potential and the relative abundance of syntrophs (Extended Data Fig. 10).

Assay I. Totally, 17 out of 67 paddy soils were used by selecting soils with relatively high proportions of *hgcA*-containing SRB, methanogens or FeRB. Detailed information on these soils and each group is listed in Supplementary Tables 1 and 6, respectively. There were nine kinds of soil with the highest relative abundance of FeRB Geobacteraceae (63.2–80.4%: HLJHEB, AHAQ, HLJQTH, GDGZ, LNDL, JXJA, HNLD, JSSZ and HLJMDJ soil, distributing in an area with a longitude of 111.38–130.59° E and latitude of 23.55–47.24° N), four types of soil with the highest relative abundances of SRB (11.9–33.9%: JSXZ, SCMY, GZZY and HBJM soil, distributing in an area within a longitude of 104.97–118.01° E and latitude of 27.73–34.34° N) and four kinds of soil with the highest relative abundance of methanogens (35.2–49.3%: HBHG, GZKL, GXHZ and GXLB, which distribute in an area with a longitude of 107.81–115.94° E and latitude of 23.73–30.07° N). For each soil, a total of four treatments (with triplicate samples per treatment) were set up: soils without adding any metabolism inhibitor (Control), soils added with the inhibitor of SRB activities (20 mM Na_2MoO_4 (ref. 13)), soils added with the inhibitor of methanogen activities (30 mM bromoethanesulfonate/BES[16] and soils added with inhibitors of both SRB and methanogens (20 mM Na_2MoO_4 and 30 mM BES). Molybdate and BES are specific inhibitors of SRB and methanogens, and their application is a traditional and acceptable method to explore the respective

role of SRB and methanogens in Hg methylation[43]. Specifically, 5 g of soil (wet weight, the same below) was resuspended in 30 ml of deionized water in a 50 ml polypropylene centrifuge tube (Corning), with Na_2MoO_4 and/or BES added or not. The tubes were then sealed and incubated at 28 °C in the dark under anoxic conditions in the anaerobic workstation (AW300SG, Electrotek), simulating bulk paddy soils during rice cultivation[13], considering that most identified microbial Hg methylators are anaerobes[9]. After 7 days, all tubes were centrifuged (2,950 × g, 20 min) to separate the soil particles and the overlying water. The soils were then determined for MeHg concentrations and HCl-extractable Fe(II) concentrations. Here, HCl-extractable Fe(II) concentrations were used to indicate iron reduction and thus FeRB activities[18], considering the predominant role of FeRB in iron reduction[44] and the minor role of Fe(II) oxidization under the experimental conditions[45], for example, pH ranging from 4.6 to 7.5 and anoxic conditions. The overlying water was measured for SO_4^{2-} (indicating sulfate reduction and thus SRB activities, using ion chromatography, Dionex, ICS1000). HCl-extractable Fe(II) in soils instead of dissolved Fe(II) in the overlying water was quantified to indicate iron reduction, as the latter fails to include solid Fe(II) species[46].

Assay II. For FeRB, no specific inhibitor of FeRB activities is available. Instead, we quantified the changes in FeRB Geobacteraceae activity (indicated by the changes in HCl-extractable Fe(II) levels) and MeHg formation in response to Se amendment, considering that either FeRB Geobacteraceae abundance or Hg methylation potential is significantly and negatively related to soil Se content on a large scale (Fig. 5 and Supplementary Discussion 2). A total of ten soil types with relatively low FeRB Geobacteraceae abundance (average 31.5%, that is, GXHZ, GZTR, HNHY and GXNN, distributing in an area with a longitude of 108.37–112.84° E and latitude of 22.82–27.69° N) and high Geobacteraceae abundance (average 76.1%, that is, AHCH, JSSZ, JXJJ, HLJHEB, HLJQTH and HLJMDJ, distributing in an area with a longitude of 114.29–130.59° E and latitude of 26.75–47.24° N) were selected. Detailed information about the soil characteristics of these two groups can be found in Supplementary Table 6. For each soil, two treatments (with triplicate samples per treatment) were set, that is, control and Se addition treatment. Specifically, 5 g of soil was resuspended in 15 ml of deionized water in a 50-ml tube. Se, in the form of Na_2SeO_3, was amended into soils at a concentration of 0.4 mg kg^{-1}, comparable to the average Se contents of these 67 soils (0.4 mg kg^{-1}). Acetate, a common carbon source for microorganisms and a main component of rice root exudates, was added to a final concentration of 10 mM to simulate the supply of carbon sources during rice cultivation[47]. After being thoroughly mixed, the soils were incubated in an artificial chamber (25 °C, dark). On the seventh day of incubation, IHg, in the form of $Hg(NO_3)_2$, was spiked to 100 μg kg^{-1}, and the Hg-spiked soils were subjected to further incubation for 24 h. Then the samples were centrifuged (2,950 × g, 20 min) and separated into soil particles and overlying water under anaerobic conditions. Afterward, the soil was divided into subsamples for the analysis of MeHg and HCl-extractable Fe(II), whereas the overlying water was subjected to sulfate analysis.

Measurement of parameters. *Hg analysis*. The measurements of soil THg were based on the EPA Method 7473 (US EPA, 1998), using the DMA-80 Direct Mercury Analyzer (Milestone). The detection limit for soil THg is 0.1 μg kg^{-1}. The MeHg analysis in soils was based on EPA 1630, which has been commonly used in previous publications[13,48]. Specifically, soils (~0.5 g) were extracted using 2 ml of 25% (w/w) KOH-CH_3OH at 70 °C for 4 h. Then, the digested sample solution was ethylated by $NaBEt_4$ with a pH of 4.1–4.5 in citrate buffer before MeHg was measured by an automatic MeHg analyser (CVAFS, Brooks Rand model III, Brooks Rand). The detection limit for soil MeHg is 0.05 μg kg^{-1}. Quality control of the Hg measurement included triplicate detections and certified reference materials (GBW07423-GSS-9, THg 32 ± 3 μg kg^{-1}; GBW07405-GSS-5, THg 290 ± 30 μg kg^{-1}; ERM-CC580, MeHg 75 ± 4 μg kg^{-1}). The recoveries of Hg ranged from 91.5% to 109.8%. Meanwhile, the formation of MeHg artefact during the extraction (with the aid of the stable isotope tracer of Hg, ^{202}Hg) and extraction efficiencies were quantified using 14 soils. Results show that the alkaline extraction procedure is well suited for analysing MeHg in paddy soils considering the neglectable artificial MeHg (<10% of ambient MeHg levels) and satisfactory recovery rates of spiked MeHg (81.4–114.6%, described in Supplementary Discussion 7).

k_m and k_d. The determinations of Hg methylation rate (k_m) and MeHg demethylation rate constant (k_d) were based on the method established by Hintelmann et al.[40]. Specifically, 2 g of soil was resuspended in 2 ml of anaerobic water (obtained via bubbling nitrogen gas in deionized water for 1 h) in a glass bottle. The bottles were sealed and incubated at 28 °C in the dark for seven days within the anaerobic workstation. After that, the overlying water and soil were separated by centrifugation (1,106 × g, 10 min). The soils were then added with $^{202}HgCl_2$ and $CH_3{}^{199}HgBr$ at 10–80% of the ambient THg or MeHg concentrations in soils, mixed thoroughly and further incubated for 24 h. Afterward, 0.7 g of soil was transferred into a 15 ml centrifuge tube, with $Me^{201}Hg$ (as internal standard) added, and digested with KOH-methanol described above. After digestion, the supernatant solutions were obtained by centrifugation and determined for $Me^{202}Hg$, $Me^{201}Hg$ and $Me^{199}Hg$ using inductively coupled plasma mass spectrometry (ICP-MS; NexION300, Perkin Elmer) coupled with a gas chromatography column. Hg methylation rate (k_m) and MeHg demethylation rate constants (k_d) are calculated as follows:

$$k_m = \frac{\left[Me^{202}Hg^+\right]_{t_f}}{\left[202_{Hg^{2+}}\right]_{t0} t} \quad (1)$$

$$k_d = \frac{\ln\left[Me^{199}Hg^+\right]_{t_0} - \ln\left[Me^{199}Hg^+\right]_{t_f}}{t} \quad (2)$$

Where $[MeHg^+]$ and $[Hg^+]$ represent the concentration of MeHg and IHg; t_0 and t_f refer to the start time and the end time and t is the incubation duration (day).

hgcA quantification. About 0.25 g of freeze-dried soil sample was prepared for total microbial DNA isolation using MoBio PowerSoil DNA Isolation Kit (QIAGEN Inc.) and the detailed process of DNA isolation was performed based on the instructions. Then, the concentration and quality of the isolated DNA were tested by a spectrophotometer (NanoDrop ND2000c UV-vis, Thermo Fisher Scientific) before storage at −20 °C.

To measure the abundances of *hgcA* genes by quantitative polymerase chain reaction (qPCR), two pairs of primers were prepared for Deltaproteobacteria and Archaea, respectively: the Oak Ridge National Laboratory (ORNL)-Delta-HgcA-F (5′-GCCAACTACAAGMTGASCWC-3′) and ORNL-Delta-HgcA-R (5′-CCSGCNGCRCACCAGACRTT-3′) for Deltaproteobacteria group with *hgcA* gene, whereas the ORNL-Archaea-HgcA-F (5′-AAYTAYWCNCTSAGYTTYGAYGC-3′) and ORNL-Archaea-HgcA-R (5′-TCDGTCCCRAABGTSCCYTT-3′) for Archaea group with *hgcA* gene[49]. Although these *hgcAB*-specific primer sets cover 64 and 86% for Deltaproteobacteria and Archaea, respectively[49], their validity for paddy soil samples has been confirmed in our previous study by comparing the results of qPCR and metagenomic sequencing analyses[12]. The 25 μl solution system for the gene amplification contained 12.5 μl of SYBR Premix Ex Taq (TaKaRa Bio Inc.), 0.5 μl of Deltaproteobacteria or Archaea forward and reverse primer (10 μM),and 2 μl of diluted (tenfold) DNA. The thermal-cycling conditions for qPCR were set as Method I (3 min of initial denaturation at 95 °C, 40 cycles of 15 s at 95 °C, 15 s at 50 °C, 15 s at 55 °C and 4 min at 72 °C, followed by a plate read at 83 °C) for Deltaproteobacteria and Method II

(3 min of initial denaturation at 95 °C, 40 cycles of 15 s at 95 °C, 30 s at 50 °C and 25 s at 55 °C, then 4 min at 72 °C, followed by a plate read at 83 °C) for Archaea, respectively[15]. Negative controls were run with each sample set and primer set using nuclease-free water and non-*hgcA*-carrying *Escherichia coli*. Plasmid DNA concentrations were quantified before each qPCR assay, and the concentration consistently corresponded to the cycle threshold (C_t) value. The qPCR products from all samples were subject to melt-curve analysis (Supplementary Fig. 1a) to evaluate potential primer-dimer formation and amplification of non-specific products. Standard curves were generated during each analytical batch using the concentration and the threshold (C_t, ranged from 15 to 35; Supplementary Fig. 1b) and had an amplification efficiency ranging from 90 to 110%. The linear correlation coefficient (R^2) of *hgcA* genes ranged from 0.997 to 1.000. The selected qPCR products were also evaluated using agarose gels to check amplicon sizes (Supplementary Fig. 1c).

hgcA-containing community analysis. The *hgcA* gene from each sample was amplified using the hgcA_261 F/hgcA_912 R primers with barcode adaptors[15]. Although potential biases might exist due to primer preference or its limitations in covering all species equally among different environmental samples, these *hgcAB*-specific primer sets are the only available tool to directly target the Hg methylation genes-carrying microbes that link to their function activities as well, and their validity has been confirmed for environmental samples[15]. Amplicons were sequenced by the Illumina MiSeq (Illumina Inc.) and pair-end 300 bp mode at Majorbio in Shanghai, China. Only the forward read sequence was used for downstream analysis due to the fragment length of the *hgcA* gene (651 bp) exceeding the available sequencing length (600 bp). The SICKLE software was used to trim the low-quality reads, then left-over adaptors and primers were removed using CUTADAPT software. A count table was generated by mapping the original quality-controlled reads to the representative sequences of the obtained clusters using the USEARCH software. The database used for annotation was based on the sequences used by Podar et al.[50]. A Hidden Markov Model was used to search the original sequence of *hgcA* amplicon sequencing for taxonomical annotation with a manually curated database of Hg-MATE-Db using HMMER.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The authors declare that the main data supporting the findings of this study are available within the article and its Supplementary Information file and the Source data. The data of rice-planting area for each country was obtained from the Food and Agriculture Organization of the United Nations (FAO) website (https://www.fao.org/faostat/en/#data/QCL). The per-capita rice consumption rates were retrieved from China Statistical Yearbook 2011 (https://www.stats.gov.cn/sj/ndsj/2011/indexch.htm). The shape data for the map of China is provided by the Resource and Environment Science and Data Center (https://www.resdc.cn/Datalist1.aspx?FieldTyepID=20,0, accessed on 18 October 2022). Source data are provided with this paper.

Code availability

Code of rice-paddy model are available in the Source data.

References

1. *Global Mercury Assessment 2018* (UNEP, 2018).
2. Kirk, L. E., Jørgensen, J. S., Nielsen, F. & Grandjean, P. Public health benefits of hair-mercury analysis and dietary advice in lowering methylmercury exposure in pregnant women. *Scand. J. Public Health* **45**, 444–451 (2017).
3. Bellanger, M. et al. Economic benefits of methylmercury exposure control in Europe: monetary value of neurotoxicity prevention. *Environ. Health* **12**, 3 (2013).
4. Rothenberg, S. E., Jackson, B. P., Carly McCalla, G., Donohue, A. & Emmons, A. M. Co-exposure to methylmercury and inorganic arsenic in baby rice cereals and rice-containing teething biscuits. *Environ. Res.* **159**, 639–647 (2017).
5. Gong, Y. et al. Bioaccessibility-corrected risk assessment of urban dietary methylmercury exposure via fish and rice consumption in China. *Sci. Total Environ.* **630**, 222–230 (2018).
6. Cui, W. et al. Occurrence of methylmercury in rice-based infant cereals and estimation of daily dietary intake of methylmercury for infants. *J. Agric. Food Chem.* **65**, 9569–9578 (2017).
7. Wang, Z., Sun, T., Driscoll, C. T., Yin, Y. & Zhang, X. Mechanism of accumulation of methylmercury in rice (*Oryza sativa* L.) in a mercury mining area. *Environ. Sci. Technol.* **52**, 9749–9757 (2018).
8. Parks, J. M. et al. The genetic basis for bacterial mercury methylation. *Science* **339**, 1332–1335 (2013).
9. Gilmour, C. C. et al. Mercury methylation by novel microorganisms from new environments. *Environ. Sci. Technol.* **47**, 11810–11820 (2013).
10. Liu, Y. R. et al. Overlooked role of putative non-Hg methylators in predicting methylmercury production in paddy soils. *Environ. Sci. Technol.* **53**, 12330–12338 (2019).
11. Beckers, F. & Rinklebe, J. Cycling of mercury in the environment: sources, fate, and human health implications: a review. *Crit. Rev. Environ. Sci. Technol.* **47**, 693–794 (2017).
12. Liu, Y.-R. et al. Unraveling microbial communities associated with methylmercury production in paddy soils. *Environ. Sci. Technol.* **52**, 13110–13118 (2018).
13. Tang, W. et al. Increased methylmercury accumulation in rice after straw amendment. *Environ. Sci. Technol.* **53**, 6144–6153 (2019).
14. Yuan, H. Y. et al. Geographic distance and amorphous iron affect the abundance and distribution of Geobacteraceae in paddy soils in China. *J. Soils Sediments* **16**, 2657–2665 (2016).
15. Bravo, A. G. et al. Geobacteraceae are important members of mercury-methylating microbial communities of sediments impacted by waste water releases. *ISME J.* **12**, 802–812 (2018).
16. Bouchet, S. et al. Linking microbial activities and low-molecular-weight thiols to Hg methylation in biofilms and periphyton from high-altitude tropical lakes in the Bolivian Altiplano. *Environ. Sci. Technol.* **52**, 9758–9767 (2018).
17. Goñi-Urriza, M. et al. Relationships between bacterial energetic metabolism, mercury methylation potential, and *hgcA*/*hgcB* gene expression in *Desulfovibrio dechloroacetivorans* BerOc1. *Environ. Sci. Pollut. Res.* **22**, 13764–13771 (2015).
18. Tong, H., Hu, M., Li, F. B., Liu, C. S. & Chen, M. J. Biochar enhances the microbial and chemical transformation of pentachlorophenol in paddy soil. *Soil Biol. Biochem.* **70**, 142–150 (2014).
19. Wang, J. et al. Mobilization, methylation, and demethylation of mercury in a paddy soil under systematic redox changes. *Environ. Sci. Technol.* **55**, 10133–10141 (2021).
20. Coleman Wasik, J. K. et al. Methylmercury declines in a boreal peatland when experimental sulfate deposition decreases. *Environ. Sci. Technol.* **46**, 6663–6671 (2012).
21. Wu, Q. et al. Methanogenesis is an important process in controlling MeHg concentration in rice paddy soils affected by mining activities. *Environ. Sci. Technol.* **54**, 13517–13526 (2020).
22. Kwon, S. Y., Selin, N. E., Giang, A., Karplus, V. J. & Zhang, D. Present and future mercury concentrations in Chinese rice: insights from modeling. *Glob. Biogeochem. Cycles* **32**, 437–462 (2018).
23. Li, C. H. et al. Pollen and phytolith analyses of ancient paddy fields at Chuodun site, the Yangtze River Delta. *Pedosphere* **17**, 209–218 (2007).

24. Liu, B. et al. Promoting potato as staple food can reduce the carbon–land–water impacts of crops in China. *Nat. Food* **2**, 570–577 (2021).
25. Ullah, H. et al. A comprehensive review on environmental transformation of selenium: recent advances and research perspectives. *Environ. Geochem. Health* **41**, 1003–1035 (2019).
26. Chen, L. et al. Determination of selenium concentration of rice in China and effect of fertilization of selenite and selenate on selenium content of rice. *J. Agric. Food Chem.* **50**, 5128–5130 (2002).
27. Wu, Q., Li, G., Wang, S., Liu, K. & Hao, J. Mitigation options of atmospheric Hg emissions in China. *Environ. Sci. Technol.* **52**, 12368–12375 (2018).
28. Lovley, D. R. et al. Geobacter. The microbe electric's physiology, ecology, and practical applications. *Adv. Microb. Physiol.* **59**, 1–100 (2011).
29. Li, X., Ding, L., Li, X. & Zhu, Y. Abundance, diversity, and structure of Geobacteraceae community in paddy soil under long-term fertilization practices. *Appl. Soil Ecol.* **153**, 103577 (2020).
30. Leclerc, M. et al. Microbial diversity and mercury methylation activity in periphytic biofilms at a run-of-river hydroelectric dam and constructed wetlands. *mSphere* **6**, e00021 (2021).
31. Yuan, K. et al. Mercury methylation-related microbes and genes in the sediments of the Pearl River Estuary and the South China Sea. *Ecotoxicol. Environ. Saf.* **185**, 109722 (2019).
32. Jones, D. S. et al. Molecular evidence for novel mercury methylating microorganisms in sulfate-impacted lakes. *ISME J.* **13**, 1659–1675 (2019).
33. Compeau, G. C. & Bartha, R. Sulfate-reducing bacteria: principal methylators of mercury in anoxic estuarine sediment. *Appl. Environ. Microbiol.* **50**, 498–502 (1985).
34. Schartup, A. T. et al. Climate change and overfishing increase neurotoxicant in marine predators. *Nature* **572**, 648–650 (2019).
35. Zhang, H. et al. Decreasing mercury levels in consumer fish over the three decades of increasing mercury emissions in China. *Eco-Environ. Health* **1**, 46–52 (2022).
36. Zhang, Y. et al. Global health effects of future atmospheric mercury emissions. *Nat. Commun.* **12**, 3035 (2021).
37. Ancora, M. P., Zhang, L., Wang, S., Schreifels, J. & Hao, J. Economic analysis of atmospheric mercury emission control for coal-fired power plants in China. *J. Environ. Sci.* **33**, 125–134 (2015).
38. Carini, P., Marsden, P. J., Leff, J. W., Morgan, E. E. & Fierer, N. Relic DNA is abundant in soil and obscures estimates of soil microbial diversity. *Nat. Microbiol.* **19**, 16242 (2016).
39. Zhao, L. et al. Mercury methylation in paddy soil: source and distribution of mercury species at a Hg mining area, Guizhou Province, China. *Biogeosciences* **13**, 2429–2440 (2016).
40. Hintelmann, H., Keppel-Jones, K. & Evans, R. D. Constants of mercury methylation and demethylation rates in sediments and comparison of tracer and ambient mercury availability. *Environ. Toxicol. Chem.* **19**, 2204–2211 (2000).
41. Fan, K., Weisenhorn, P., Gilbert, J. A. & Chu, H. Wheat rhizosphere harbors a less complex and more stable microbial co-occurrence pattern than bulk soil. *Soil Biol. Biochem.* **125**, 251–260 (2018).
42. Yu, R. Q., Reinfelder, J. R., Hines, M. E. & Barkay, T. Syntrophic pathways for microbial mercury methylation. *ISME J.***12**, 1826–1835 (2018).
43. Lei, P. et al. Properly interpret metabolic inhibition results to identify primary mercury methylating microbes. *Crit. Rev. Environ. Sci. Technol.* **53**, 1757–1773 (2023).
44. Lovley, D. R., Holmes, D. E. & Nevin, K. P. Dissimilatory Fe(III) and Mn(IV) reduction. *Adv. Microb. Physiol.* **49**, 219–286 (2004).
45. Balci, N. et al. Iron isotope fractionation during microbially stimulated Fe(II) oxidation and Fe(III) precipitation. *Geochim. Cosmochim. Acta* **70**, 622–639 (2006).
46. Ratering, S. & Schnell, S. Localization of iron-reducing activity in paddy soil by profile studies. *Biogeochemistry* **48**, 341–365 (2000).
47. Kerin, E. J. et al. Mercury methylation by dissimilatory iron-reducing bacteria. *Appl. Environ. Microbiol.* **72**, 7919–7921 (2006).
48. An, J. et al. Mercury uptake by *Desulfovibrio desulfuricans* ND132: passive or active? *Environ. Sci. Technol.* **53**, 6264–6272 (2019).
49. Christensen, G. A. et al. Development and validation of broad-range qualitative and clade specific quantitative molecular probes for assessing mercury methylation in the environment. *Appl. Environ. Microbiol.* **82**, 6068–6078 (2016).
50. Podar, M. et al. Global prevalence and distribution of genes and microorganisms involved in mercury methylation. *Sci. Adv.* **1**, e1500675 (2015).

Acknowledgements

We appreciate G. Qiu (Institute of Geochemistry, Chinese Academy of Sciences) for providing data on site-specific rice MeHg concentrations for the model evaluation. W. Tang appreciates the financial support from National Natural Science Foundation of China (42107223), Fundamental Research Funds for the Central Universities (0211-14380175) and Natural Science Foundation of Jiangsu Province (BK 20230082 and BK 20190319). H.Z. gets supports from National Natural Science Foundation of China (41673075). Y.-R.L. acknowledges financial support from National Natural Science Foundation of China (42177022), and L.C. appreciates National Natural Science Foundation of China (41701589). H.R. is supported by the financial support from National Natural Science Foundation of China (52388101).

Author contributions

H.Z., Y.-R.L., L.C. and W. Tang conceived and designed this study. W. Tang, Z.L. and Y.H. conducted all the experiments. Y.-R.L. led the analysis of microbial community. Z.L. and X.Z. modified the RPBM with the help from S.Y.K. L.C. and L.M.N. did the sensitivity analysis. H.Z., Y.-R.L., L.C. and W. Tang led the manuscript writing. C.S. and M.W. significantly contributed to the writing of the manuscript, and other authors participated in the writing.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s43016-024-00954-7.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s43016-024-00954-7.

Correspondence and requests for materials should be addressed to Huan Zhong, Christian Sonne, Long Chen or Yu-Rong Liu.

Peer review information *Nature Food* thanks Shaojie Song, Yuebing Sun, Anil Somenahally and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

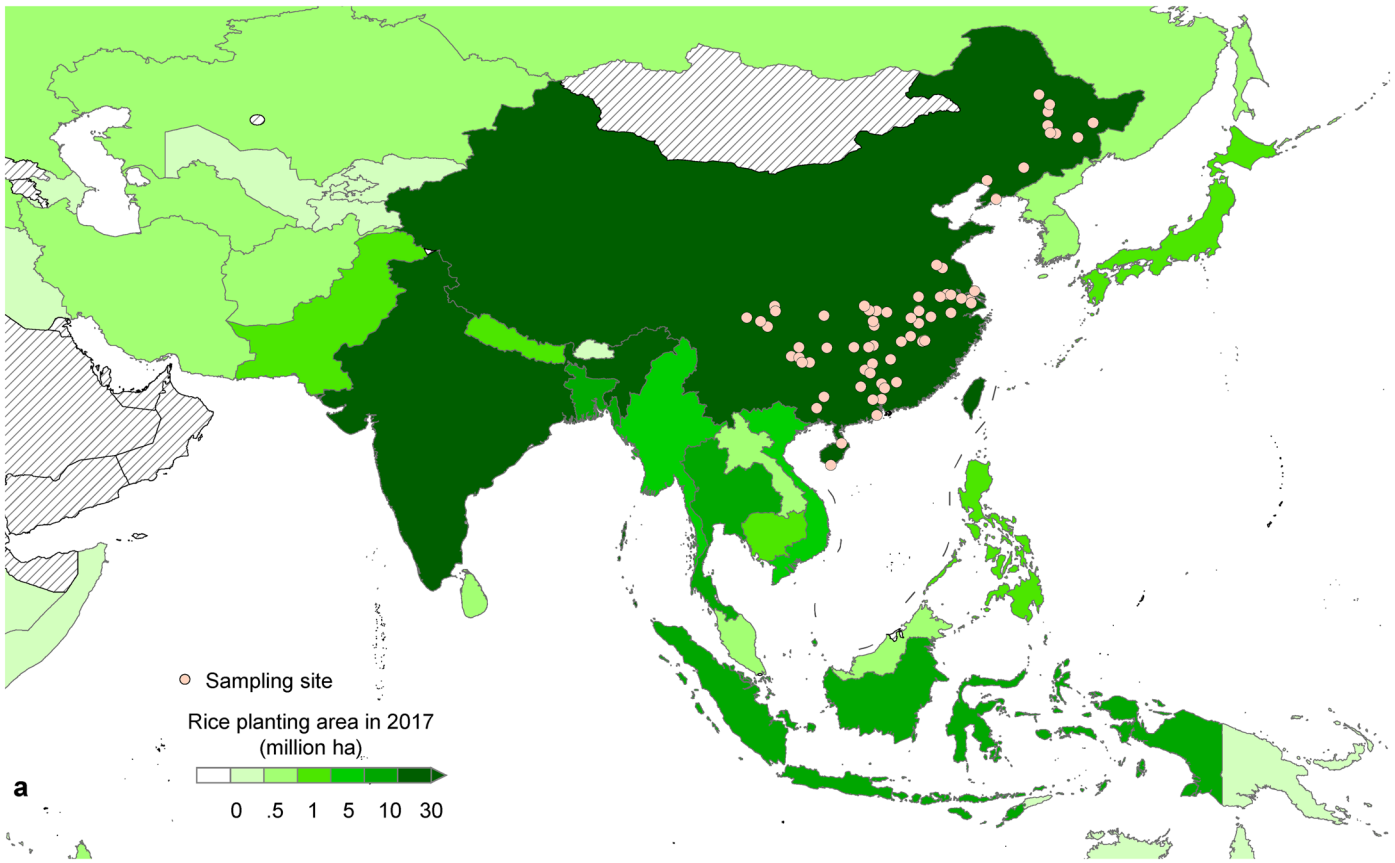
Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

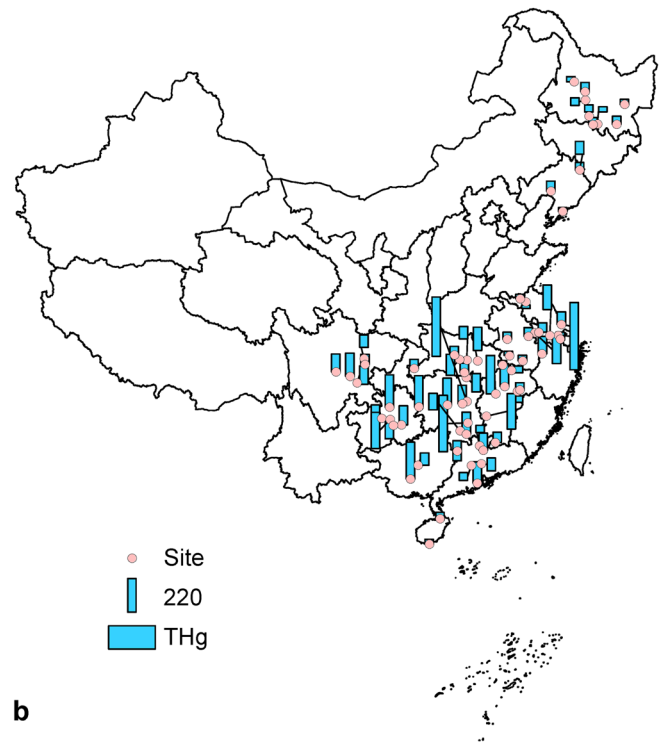
Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

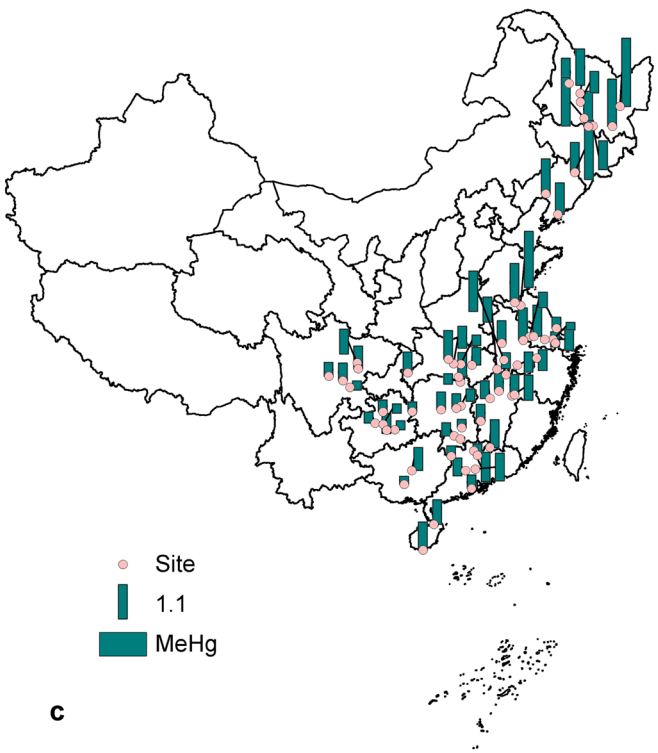
© The Author(s), under exclusive licence to Springer Nature Limited 2024

[1]School of the Environment, Nanjing University, State Key Laboratory of Pollution Control and Resource Reuse, Nanjing, China. [2]Environmental and Life Sciences Program (EnLS), Trent University, Peterborough, Ontario, Canada. [3]Department of Ecoscience, Aarhus University, Roskilde, Denmark. [4]Sustainability Cluster, School of Engineering, University of Petroleum and Energy Studies, Dehradun, India. [5]Higher Institution Centre of Excellence (HICoE), Institute of Tropical Aquaculture and Fisheries (AKUATROP), Universiti Malaysia Terengganu, Kuala Nerus, Malaysia. [6]Center for Global Health Research (CGHR), Saveetha Medical College, Saveetha Institute of Medical and Technical Sciences (SIMATS), Saveetha University, Chennai, India. [7]Division of Environmental Science and Engineering, Pohang University of Science and Technology, Pohang, South Korea. [8]School of Architecture and Civil Engineering, Institute of Foundation Engineering, Water and Waste Management, Laboratory of Soil and Groundwater Management, University of Wuppertal, Wuppertal, Germany. [9]Faculty of Sciences and Technology, Civil Engineering Research and Innovation for Sustainability Center, University of Algarve, Faro, Portugal. [10]Department of Biology, Center for Environment, Biodiversity and Conservation, The University of Texas at Tyler, Tyler, TX, USA. [11]Environmental Sciences Division, Oak Ridge National Laboratory, Oak Ridge, TN, USA. [12]Department of Chemistry, Trent University, Peterborough, Ontario, Canada. [13]School of Life Sciences, Earth and Environmental Sciences Programme, Institute of Environment, Energy and Sustainability, The Chinese University of Hong Kong, Hong Kong SAR, China. [14]State Key Laboratory of Marine Pollution, City University of Hong Kong, Hong Kong SAR, China. [15]Department of Environmental Science, Zhejiang University, Hangzhou, China. [16]National Key Laboratory of Agricultural Microbiology and College of Resources and Environment, Huazhong Agricultural University, Wuhan, China. [17]Key Laboratory of Geographic Information Science (Ministry of Education), School of Geographic Sciences, East China Normal University, Shanghai, China. [18]School of Water Resources and Environment, MOE Key Laboratory of Groundwater Circulation and Environmental Evolution, China University of Geosciences (Beijing), Beijing, China. [19]These authors contributed equally: Huan Zhong, Wenli Tang, Zizhu Li. ✉e-mail: zhonghuan@nju.edu.cn; cs@bios.au.dk; chenlong@geo.ecnu.edu.cn; yrliu@mail.hzau.edu.cn

https://doi.org/10.1038/s43016-024-00954-7

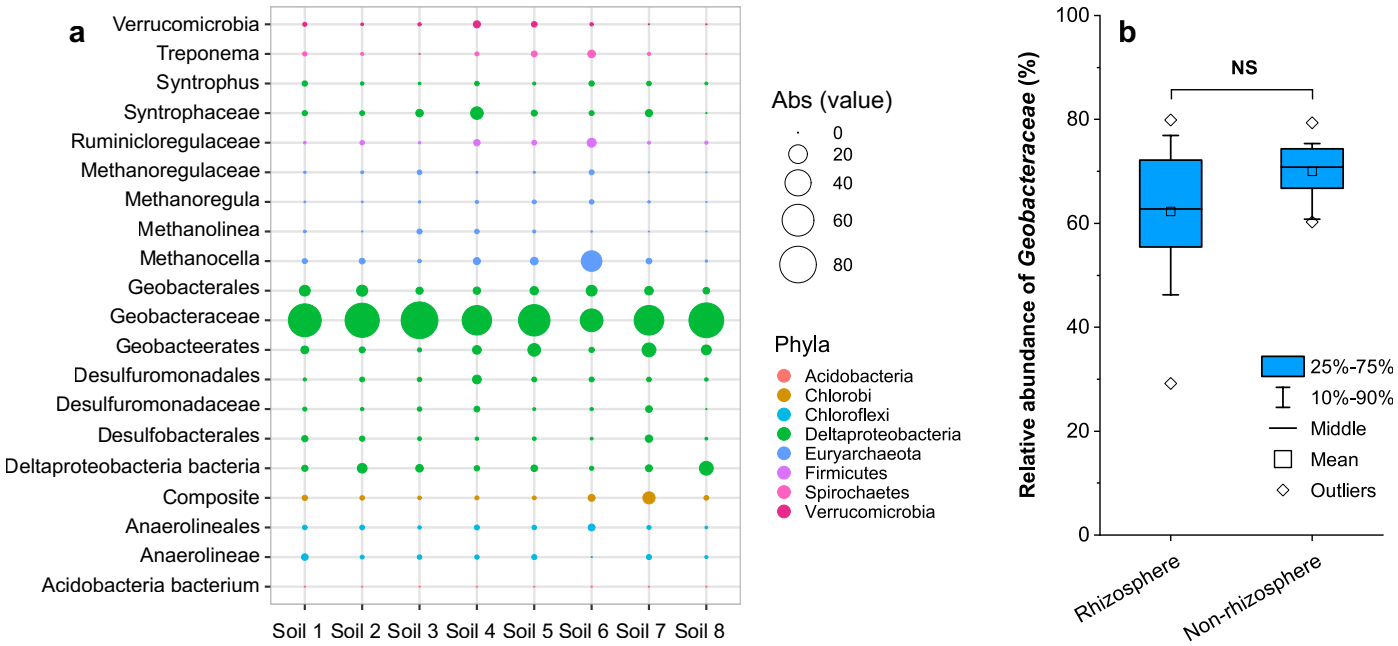






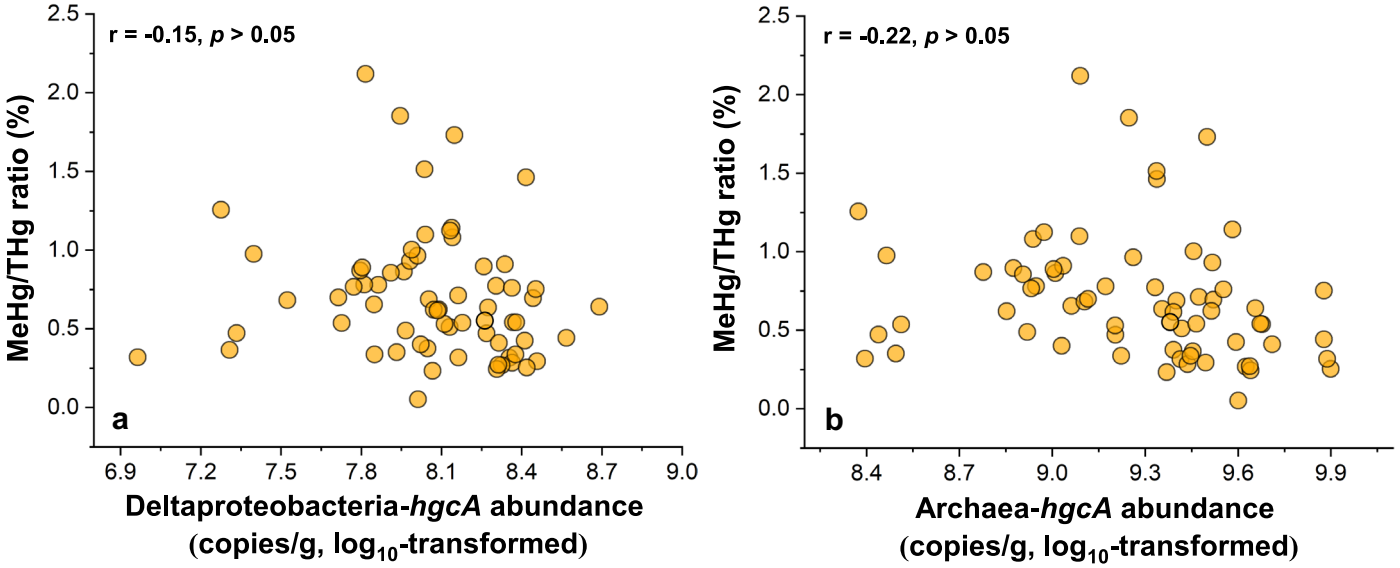
Extended Data Fig. 1 | Location, rice yield (annual provincial yield, a), and Hg concentrations (b and c) in 67 sites of paddy soil across Asia. The background colors in panel (a) indicate the national rice planting areas (million ha, 2017) in these countries and these data were obtained from the FAO website (accessed on Oct. 14 2022). The global map is provided by Natural Earth (https://www.naturalearthdata.com/downloads/10m-cultural-vectors/, accessed on Oct. 18 2022). The base map of China is provided by the Resource and Environment Science and Data Center (https://www.resdc.cn/Datalist1.aspx?FieldTyepID=20,0, accessed on 18 October 2022).

https://doi.org/10.1038/s43016-024-00954-7



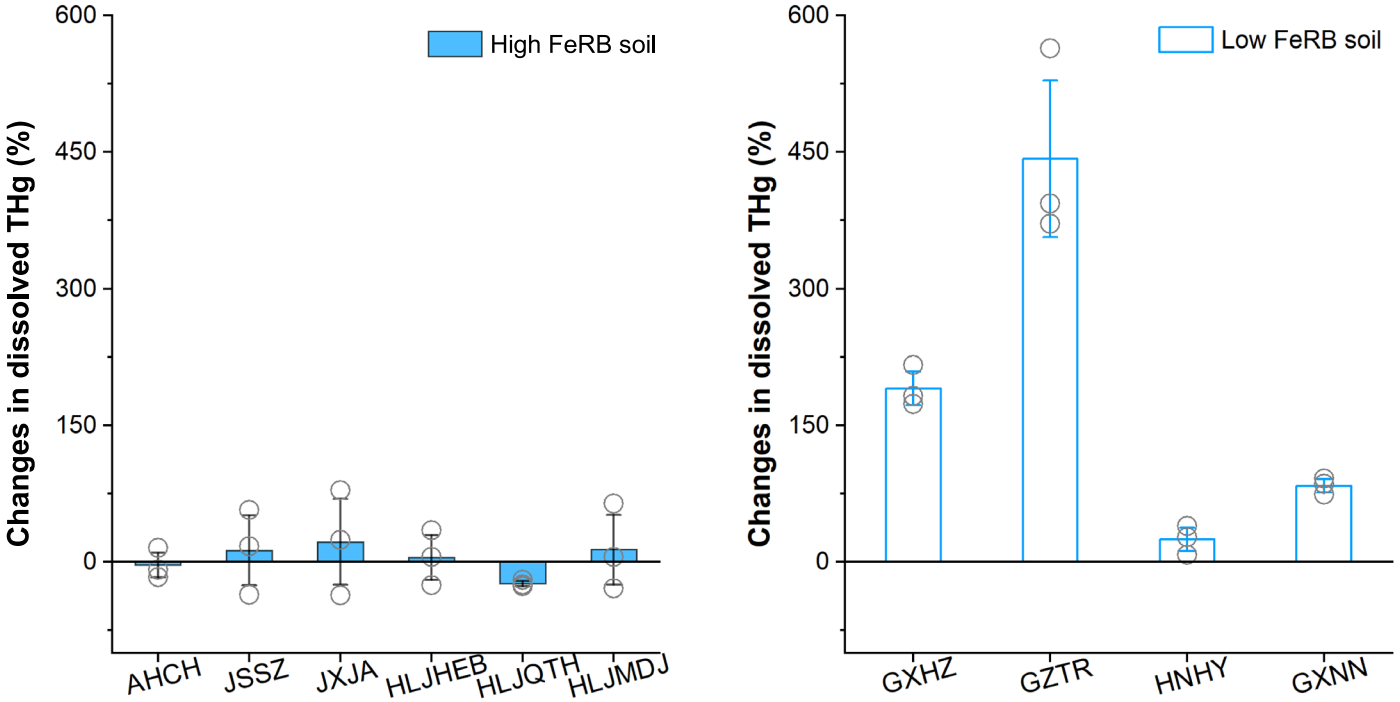
Extended Data Fig. 2 | Microbial community of *hgcA*$^+$ microbes in rice paddy soils during rice cultivation (a) and the relative abundance of *hgcA*-containing *Geobacteraceae* in rhizospheric and non-rhizospheric paddy soils (b). NS indicates there are no significant ($p > 0.05$; one-way ANOVA) differences between the two groups. N = 15 for each type of soils. For box plot, the box, whisker, line, square, and diamond indicate a 25–75% range, 10–90% range, median, mean, and outliers, respectively.

https://doi.org/10.1038/s43016-024-00954-7



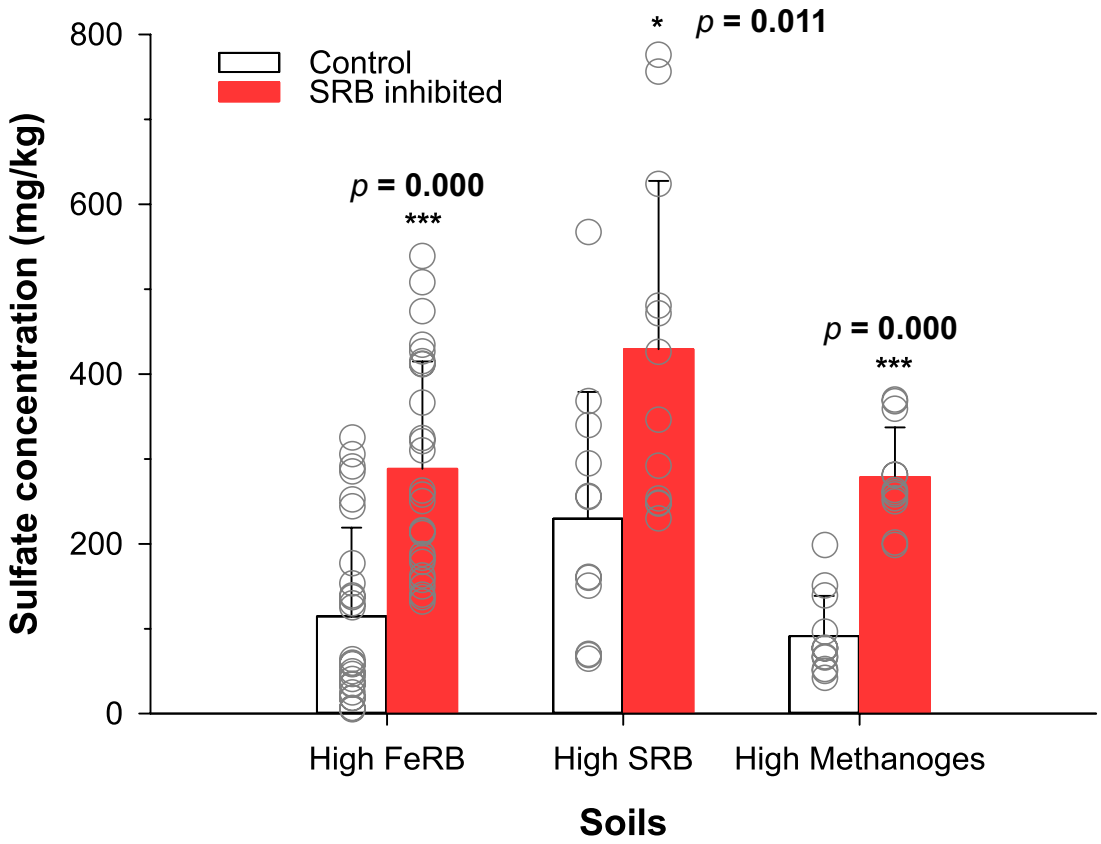
Extended Data Fig. 3 | The relationship between Hg methylation potential (the ratio of soil MeHg to soil THg, MeHg/THg, %) and *hgcA*-containing Deltaproteobacteria (a) or Archaea (b). *P* values are obtained based on a two-sided test.

https://doi.org/10.1038/s43016-024-00954-7



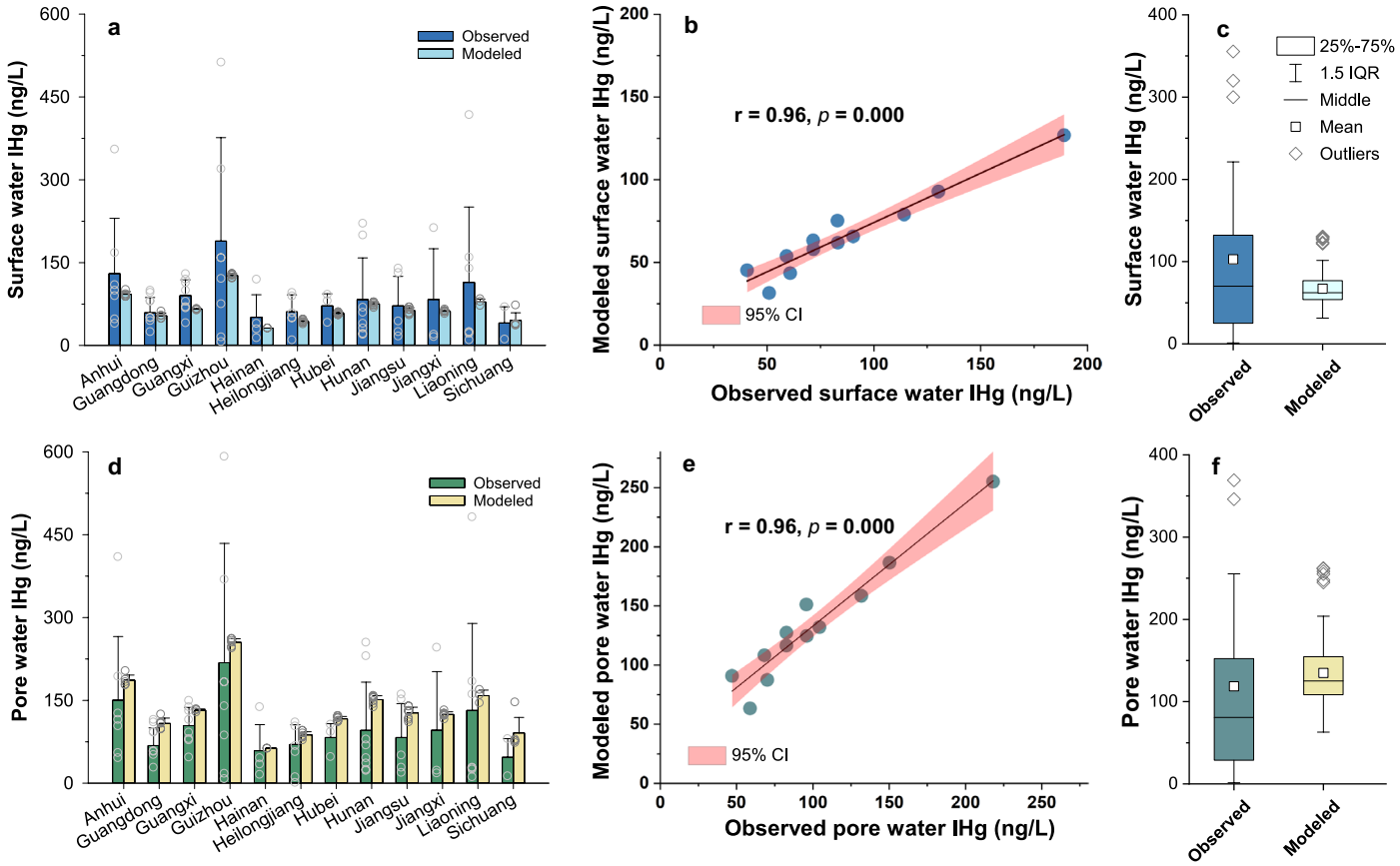
Extended Data Fig. 4 | Changes in dissolved THg concentration in the overlying water in the metabolism inhibition assays with Se addition. Extended Data Fig. 4 Data are presented as mean ± SD, n = 3 replicates.

https://doi.org/10.1038/s43016-024-00954-7



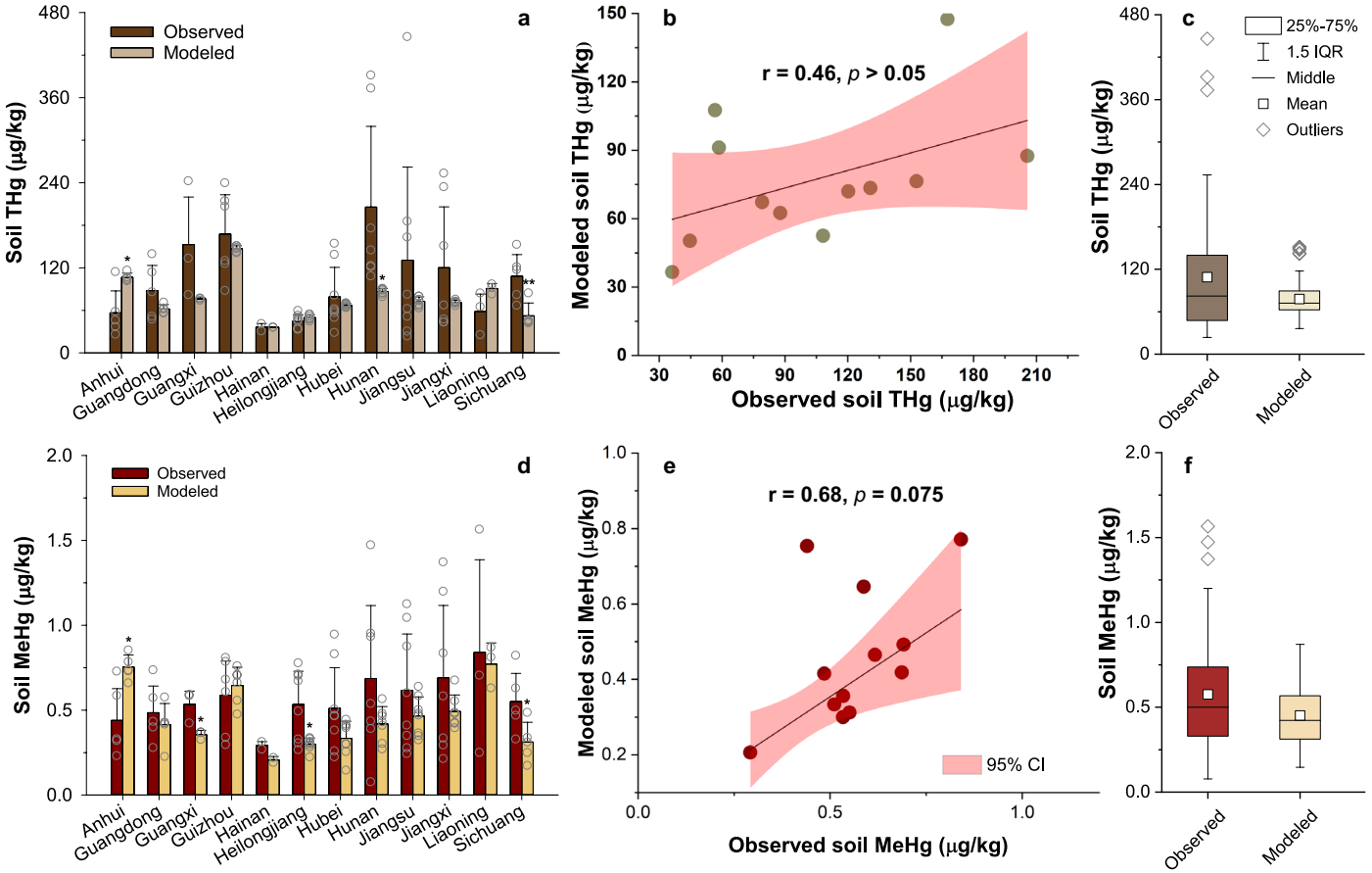
Extended Data Fig. 5 | The concentrations of sulfate in metabolism inhibition assay I. Asterisk indicates the significant difference (*, $p < 0.05$; ***, $p < 0.001$; one-way ANOVA) between two treatments. Data are presented as mean ± SD. For high FeRB soils, n = 9 soils × 3 replicates for each soil. For high SRB and high methanogens soil, n = 4 soils × 3 replicates for each soil. The value obtained from each replicate is plotted as a dot on the bar.

https://doi.org/10.1038/s43016-024-00954-7



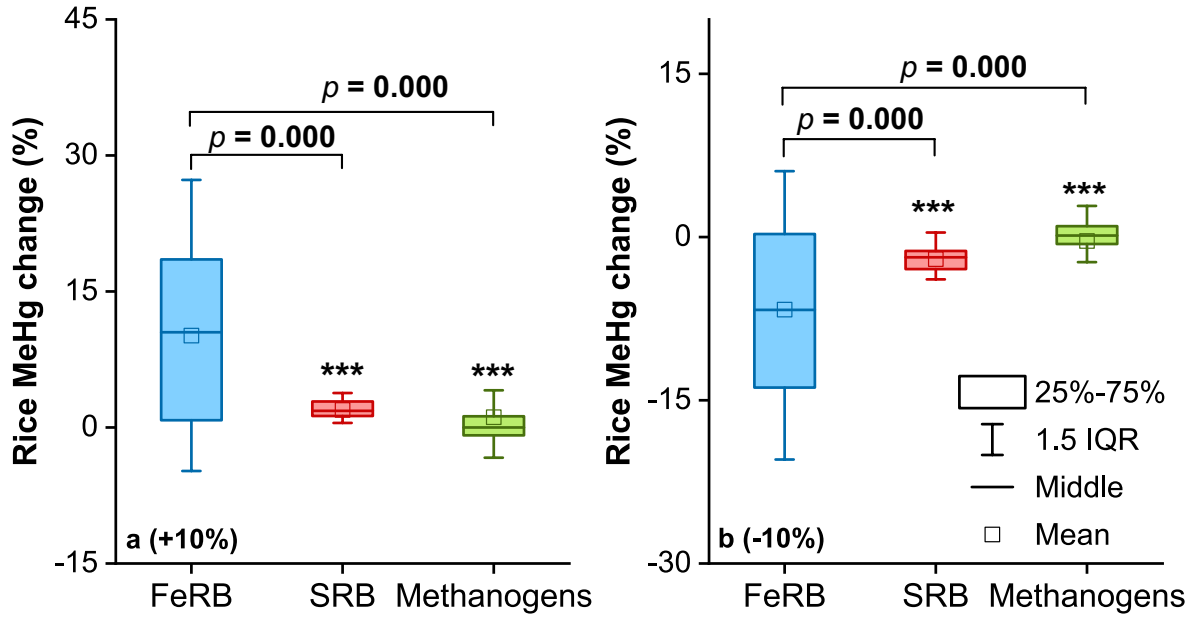
Extended Data Fig. 6 | The evaluation of the THg concentration in surface water (a, b, c) and pore water (d, e, f). In panels (a) and (d), data are presented as mean ± SD, n = 7 (Anhui), 10 (Guangdong), 8 (Guangxi), 16 (Guizhou), 4 (Hainan), 5 (Heilongjiang), 3 (Hubei), 8 (Hunan), 5 (Jiangsu), 3 (Jiangxi), 7 (Liaoning), and 2 sites (Sichuan) for the observed values, and n = 5, 5, 3, 6, 2, 8, 8, 7, 8, 7, 3, 5 sites for the modeled values for above provinces. Each dot on the bar indicates the value for a site. In panels (b) and (e), each dot represents a provincial average. *P* values are obtained based on a two-sided test. Error bands are 95% confidence intervals of the regression lines. In panels (c) and (f), the box, whisker, line, square, and diamond indicate a 25–75% range, 1.5 IQR range, median, mean, and outliers, respectively. The sample size is 78 and 67 sites for the Observed and Modeled, respectively.

https://doi.org/10.1038/s43016-024-00954-7



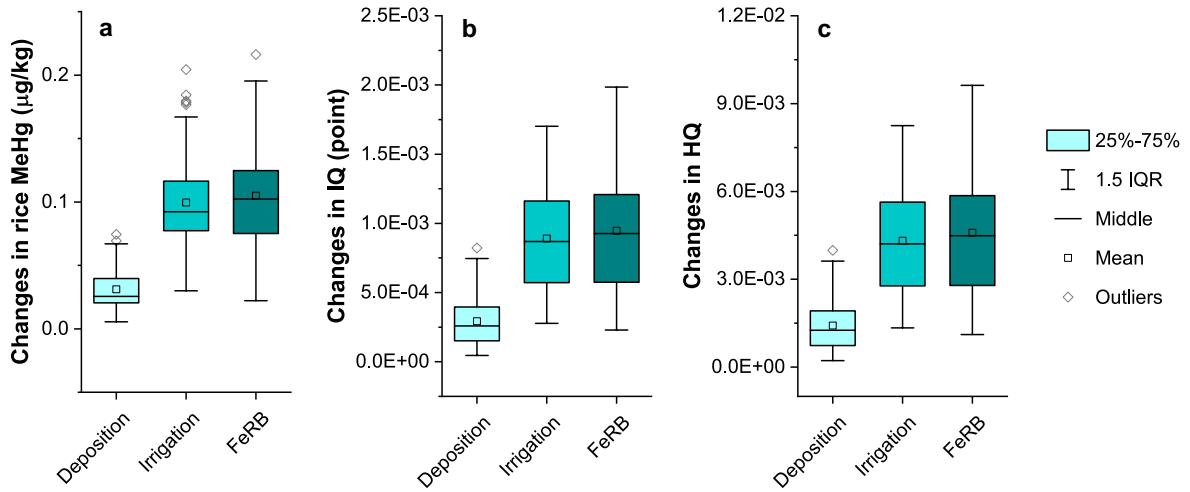
Extended Data Fig. 7 | The evaluation of the THg (a, b, c) and MeHg concentration in soils (d, e, f). In panels (a) and (d), data are presented as mean ± SD, n = 7 (Anhui), 10 (Guangdong), 8 (Guangxi), 16 (Guizhou), 4 (Hainan), 5 (Heilongjiang), 3 (Hubei), 8 (Hunan), 5 (Jiangsu), 3 (Jiangxi), 7 (Liaoning), and 2 sites (Sichuan) for the observed values, and n = 5, 5, 3, 6, 2, 8, 8, 7, 8, 7, 3, 5 sites for the modeled values for above provinces. Each dot on the bar indicates the value for a site. Asterisk indicates the significant difference (*, $p < 0.05$; **, $p < 0.01$; one-way ANOVA) between the observed and modeled values. In panels (b) and (e), each dot represents a provincial average. *P* values are obtained based on a two-sided test. Error bands are 95% confidence intervals of the regression lines. In panels (c) and (f), the box, whisker, line, square, and diamond indicate a 25–75% range, 1.5 IQR range, median, mean, and outliers, respectively. The sample size is 78 and 67 sites for the Observed and Modeled, respectively.

https://doi.org/10.1038/s43016-024-00954-7



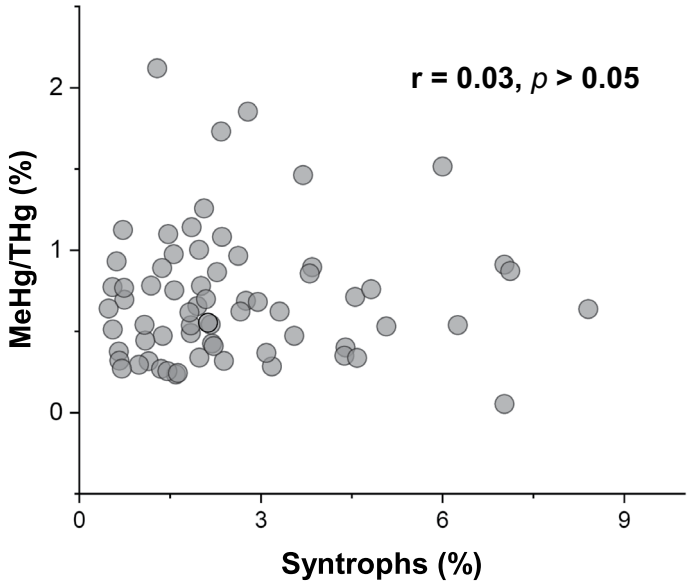
Extended Data Fig. 8 | The responses of rice MeHg to changes in the relative abundance of FeRB, SRB, or methanogens. (**a**) increase of 10% in the relative abundance of microbial methylators. (**b**) decrease of 10% in the relative abundance of microbial methylators. Asterisk indicates the significant difference (***, $p < 0.001$; one-way ANOVA). The box, whisker, line, and square indicate a 25–75% range, 1.5 IQR range, median, and mean values, respectively. The sample size is 67 sites for model simulations.

https://doi.org/10.1038/s43016-024-00954-7



Extended Data Fig. 9 | The responses of rice MeHg (a), changes in IQ decrement (b) and HQ (c) to a 10% decrease in deposition, irrigation, or FeRB abundance. The box, whisker, line, square, and diamond indicate a 25–75% range, 1.5 IQR range, median, mean, and outliers, respectively. The sample size is 67 sites for model simulations.

https://doi.org/10.1038/s43016-024-00954-7



Extended Data Fig. 10 | Relationships between Hg methylation potential in soil and the relative abundance of *Syntrophs*. *P* values are obtained based on a two-sided test.

nature portfolio

Corresponding author(s): Huan Zhong, Christine Sonne, Long Chen, and Yu-Rong Liu

Last updated by author(s): Feb 28, 2024

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐	A description of all covariates tested
☒	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	GEOS-Chem chemical transport model 9.2, GMAO Goddard Earth Observing System (GEOS-5)
Data analysis	SICKLE, CUTADAPT, USEARCH, Arcmap 10.8., Matlab R2022a, HMMER, PASW Statistics 18.0, Sigma plot 14.0, Origin 2018, Adobe Illustrators 2022

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

The authors declare that the main data supporting the findings of this study are available within the article and its Supplementary Information file. Raw data and Matlab code are provided as Source Data. The data of rice planting area for each country was obtained from the FAO website (https://www.fao.org/faostat/en/#data/QCL). The per-capita rice consumption rates was retrieved from China Statistical Yearbook 2011 (https://www.stats.gov.cn/sj/ndsj/2011/indexch.htm).The base map of China is provided by the Resource and Environment Science and Data Center (https://www.resdc.cn/Datalist1.aspx?FieldTyepID=20,0, accessed on 18 October 2022).

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	Not applicable.
Reporting on race, ethnicity, or other socially relevant groupings	Not applicable.
Population characteristics	Not applicable.
Recruitment	Not applicable.
Ethics oversight	Not applicable.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	This study analyzed the microbial composition of Hg methylators in 67 paddy soils across China and revealed that Geobacteraceae are the most abundant methylators playing a dominant role in MeHg production. By applying a rice paddy biogeochemical model, we found out that Geobacteraceae can also predict MeHg accumulation in rice. Regulating Geobacteraceae activity or abundance, e.g., via Se fertilization, is a cost-effective approach to mitigate Hg risks.
Research sample	Soil samples collected from paddy field during the period of rice growth (8 groups) and after the rice harvest (67 sites).
Sampling strategy	67 kinds of soils were collected to obtain the compositions of microbial Hg methylators in soils vary in soil properties and distribute in a large area (overing tropic, subtropic, and temperate zones in the northern hemisphere). 6 paddy soil samples were collected at the tillering stage from Hubei province and 2 paddy soils were collected at the filling stage in Jiangsu Province, China., to explore the effects of rice cultivation on the composition of microbial Hg methylators. Additional 5 groups of paired rhizospheric and bulk soils were collected from Hubei Province right after the rice harvest to reveal the differences in the relative abundance of FeRB in these soils. Soils were collected with a depth from 1-13 cm to obtain active microorganisms in paddy soils.
Data collection	Zizhu Li dominated the soil collection, with the help from Xin-Quan Zhou and all the members from Prof. Huan Zhong's team in Nanjing University. Zizhu Li, Wenli Tang , and Yunyun Hao collected all the experimental data. Zizhu Li, Wenli Tang and Xiao Zhang collected all the data used for the model evaluation. Wenli Tang conducted the cost estimation.
Timing and spatial scale	Sample collection was began in November 2017 and finished in October 2020, and these samples distributed with latitude ranging from 102.74 °N to 130.59 °N and longitude ranging from 18.26 °E to 48.04 °E, covering tropic, subtropic, and temperate zones in the northern hemisphere.
Data exclusions	No data were excluded.
Reproducibility	Key experiments were repeated by different persons and the results (e.g., metabolism inhibition assay) are consistent.
Randomization	Monte Carlo simulation is used to quantify the contributions of Hg input and FeRB abundance to MeHg accumulation in rice.
Blinding	Not applicable as this is an observational study.

Did the study involve field work? ☐ Yes ☒ No

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	Not applicable.
Novel plant genotypes	Not applicable.
Authentication	Not applicable.